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Forecasting the Success of Environmental and Sustainability Activities in International Development Using Language Models

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Abstract

Abstract in English

International aid and cooperation improve the lives of the world's poorest in developing countries act to safeguard the environment and materially promote sustainability in these countries. However, international aid activities sometimes fail to achieve their objectives. There have been few attempts in the literature to create models to explicitly predict the success of international aid activities, and none focus solely on environmental outcomes. This thesis produces a forecasting system for various metrics measuring the success of international aid activities at time of evaluation from the International Aid and Transparency Initiative (IATI) database, combining classical statistical methods with modern language model techniques. Novel techniques are applied to improve forecasting skill, including using the reasoning abilities and information gathering ability of large language models (LLMs) to improve forecasts, introducing LLM summaries of various dimensions of activity documents, and defining a novel narrative similarity grade benchmark. Overall success ratings on a scale from 1 to 6 are first assessed, then narrative forecast accuracy is measured, and finally cost-effectiveness forecasting is assessed. While no methods had consistently high accuracy in forecasting overall success, statistical models could reliably rank above-chance which activities are more likely to succeed for activities with the same reporting organization and start year ("pairwise ranking"). The full forecasting system outperforms what could be extrapolated from the stated risks in activity documents alone for overall evaluations in tests against a held-out test set of 200 latest-starting evaluated activities in a dataset of 1,181 environmental and sustainability improving activities restricted to 4 reporting organizations. Compared to the baseline of 51 % pairwise ranking for the risks extrapolation, the chosen statistical model with all input features reached 60 % [95% CI: 53 %, 66 %] on the held-out set. Across 14 binary outcome tags (covering activity rescoping, finance and budget, and target achievement), statistical models achieve an average pairwise ranking of 59 % (range: 47 %–71 %) for the 45% of pairs with differing outcomes averaging two chosen methods on the test set. This work lays the foundation to improve decision making for a wide range of initiatives and policies in developing countries and also in other data-rich institutional contexts.

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1 Introduction

1.1 Research Questions

This thesis analyzes the applicability and composability of machine learning techniques in forecasting the success of international aid activities using activity information available at the approval stage. The standard approach in the literature for assessing aid impact and success in environmental domains has historically involved exclusively "statistical forecasting": econometric techniques, linear regression models, and occasionally modern nonlinear models such as Random Forest (RF) or XGBoost (Goldemberg, Jordan, and Kenyon, 2025). However, a thorough assessment of methods to forecast activity success in this domain has yet to be conducted. In addition to statistical forecasting, the use of Large Language Model (LLM) implemented "judgemental forecasting" is proposed (Halawi et al., 2024 | "Approaching Human-Level Forecasting with Language Models"). Judgemental forecasting allows forecasters to use tools including Fermi estimates, intuition, and information gathering to make calibrated predictions. It is particularly appropriate when explicit reasoning can improve outcomes beyond what purely statistical correlations reveal.

The modelling system developed in this work predicts how successful interventions will be in the context of international aid activities affecting the environment, utilizing statistical forecasting, judgemental forecasting, and a mixture of the two. Metadata and PDF files from thousands of activities were obtained, and each record was separated into information about the activity available at approval and evaluation information about how successful the activity was. This data is used to answer the following research questions:

1. Can the overall rating, common true/false activity outcomes, or activity cost-effectiveness be forecasted from pre-activity information alone?

2. How do judgemental forecasting methods compare to statistical models in forecasting international aid overall success ratings and common true/false outcome probabilities?

3. How do differing methods of combining judgemental and statistical forecasting compare in this domain?

4. What methods improve the accuracy of narrative (qualitative) forecasts in this domain?

5. What aspects of the activity available in the dataset at the beginning of the activity lead to higher or lower overall ratings or true/false outcome probabilities?

1.2 Evaluating Activity Forecasts

To answer the above research questions, activity outcomes are assessed using four separate evaluative approaches.

First, the overall evaluation rating, a numerical score assigned by the funding body at the time of evaluation, serves as the primary quantitative outcome. Second, narrative forecast similarity grades evaluate how closely an LLM-generated qualitative prediction matches the post-activity evaluation summary, providing a measure of forecast quality that is harder to game than ratings alone. Third, aggregate cost-effectiveness Z-scores, derived from quantitative results extracted from evaluation documents, offer a proxy for the relative cost-effectiveness between activities. Fourth, in addition to quantitative cost-effectiveness outcomes and overall ratings, 14 separate true/false outcome tags pertaining to common activity outcomes were forecasted.

Overall ratings are a standardized, commonly reported, and commonly assessed metric for international aid activities. A large literature of understanding patterns in overall ratings exists. These advantages to forecasting this metric are offset by the use of overall ratings as a determination of whether future funding should continue for similar programs. As a result, overall ratings are subject to "gaming", where the rating is increased despite the activity not meriting that outcome (Goldemberg, Jordan, and Kenyon, 2025). Such gaming is inconsistent across activities and thus motivates further work to use more comparable metrics.

Simply assessing whether a forecast of overall rating prediction is accurate may be misleading, as more accurate forecasts can be correct for the wrong reasons and thus not be helpful to aid decision makers. The LLM models output overall forecasts of how an activity will proceed in narrative form, and thus can be qualitatively assessed against the narrative outcome summaries of evaluation documents. This provides an informative perspective on how differing methods to gather information pre-activity ultimately influence LLM forecasts.

One alternative to overall ratings is aggregate outcome cost-effectiveness. By extracting the quantitative achievements of activities and dividing by the expenditure, an approximation of the cost-effectiveness can be made that is roughly comparable between activities. This can be averaged with other indicators of cost-effectiveness such as reported benefit to cost ratio, or economic internal rates of returns. Because quantitative achievements divided by expenditure are not often a key criterion for future funding, such a metric is less likely to be gamed. Cost-effectiveness is also less subjective than an evaluator's overall rating, and is less susceptible to systematic differences between evaluators.

While cost-effectiveness provides useful alternative to ratings that are less likely to be gamed, accurate extraction of quantitative outcomes per activity and total expenditure are

challenging to extract reliably, suffering from lack of consistent reporting and definitions of quantitative units, differing choices of which proxy for cost-effectiveness are used, and differing definitions of what the total expenditure of the project are (such as whether expenditure is counted from only the reporting organization, or additional funding from recipient governments).

The fourth metric to assess overall activity success was a set of 14 extracted true/false outcome tags. Unlike overall rating and cost-effectiveness, the forecast is probabilistic in nature. Outcome tags describe specific, discrete events or conditions that either occurred or did not occur in a project, such as whether the closing date was extended, whether targets were met, or whether funds were reallocated. Unlike the scalar overall success rating, outcome tags capture qualitatively distinct axes of project execution and make it possible to forecast specific types of project behavior rather than a single aggregate score. Thus, outcome tags provide a richer, more interpretable characterization of what happened in each activity, making the forecasting system substantially more useful for aid evaluators than a single rating alone. Because several outcome tags were trained independently, claims about methodological improvements, feature usefulness, and feature importance ordering can all be stated much more confidently if the improvement occurs across several separate tags. Outcome tags are as a whole objective, generally clearly comparable between activities, and do not suffer from differing definitions as much as the prior two metrics. However, some outcome tags may not always be mentioned in outcome summaries.

1.3 Prior Work

An extensive literature review was undergone prior to beginning work on this thesis. A result of this review is that only one recent academic work can be said to be clearly comparable to this thesis: (Goldemberg, Jordan, and Kenyon, 2025). To the knowledge of the author, only (Goldemberg, Jordan, and Kenyon, 2025) went beyond simply evaluating overall activity ratings, which is the core focus of the majority of existing aid evaluation statistical models to actual outcome forecasing. Additionally, only (Goldemberg, Jordan, and Kenyon, 2025) uses nonlinear techniques to forecast predictions, rather than linear techniques as the rest of the literature does. Lastly, only (Goldemberg, Jordan, and Kenyon, 2025) utilize LLMs and language model embeddings from prior literature in the domain of international aid. Their work identified ‘embeddings” of LLMs were the most important features used to predict country-level sector outcomes from the World Bank World Development Indicators.

A core focus of this thesis is forecasting, which is notably absent from aid evaluation literature to the knowledge of the author. While many works in the literature have attempted to assess correlations between quantitative features and aid evaluation ratings,

they have not focused on what knowledge would be available at approval for the activity, nor do they assess out-of-time generalization of these correlations. Out-of-time generalization refers to model performance on activities that started after those used for training, as opposed to out-of-distribution generalization, which refers to performance on activities from different domains or data sources. Out-of-time prediction is critical because in order to improve aid decision making, one must assess the ability of models to forecast the outcomes of future interventions, not simply statically analyze a corpus of past activities.

While the methodology thus appears to be novel within the specific domain of international aid evaluation, it has precedence in other related academic domains.

Within the domain of LLM use, there have been early attempts at using LLMs to improve decision making for the environment. A recent tool called "climsight" summarizes and aggregates information about climate adaptation and mitigation (Koldunov and Jung, 2024), but stops short of making forecasts towards adaptation. Machine learning and LLMs have been used to collect over 80,000 articles about climate adaptation and provide analysis about which areas of implementation are lacking and point out gaps in attention towards promising categories of policies (Callaghan et al., 2025).

Limited work has also been done using LLMs such as GPT-4 to serve as data sources for policy deliberation and multi-criteria assessment of climate and sustainability interventions, finding that GPT-4 is in rough agreement with the policy rankings of human experts for the expected outcomes (Bina et al., 2025). However, very little is done to improve on GPT-4's abilities, the assessment was made on only a few dozen generic policy examples, and no attempt was made to compare outcomes between these policies and real-world outcomes. Despite these limitations, the findings are promising. For multiple criteria decision making (MCDM), GPT-4 provided a useful collaborative starting point, eased the process of considering multiple criteria effectively, and aided policy deliberation on climate change and sustainability.

2 Methods

The methods section introduces several methods to improve forecasting performance. Methods were split into simple "baseline" methods, and compared to more sophisticated "experimental" methods to ensure that the experimental methods adopted improve forecasting performance above their respective baseline methods on the held-out test set. A summary of the methods can be found at the end of this chapter in section 2.8.

2.1 Temporal Splits and Information Leakage

Train activities started on or before February 6, 2013; Validation activities started between February 7, 2013 and June 6, 2016; Test activities started after June 6, 2016 and before January 1, 2020. These date boundaries are chosen to yield approximately 700 training activities, 300 validation activities, and 200 test activities for overall ratings. See Figure 2 for the number of activities per year in each and Figure 4 for the average duration of activities within each split.

A forecasting system designed to predict the future must not gain any information that could not be known at the start of the international aid activity. The methodology was designed such that that no information from the test set was used for model selection and validation, in order to provide a realistic assessment of the forecasting skill of the model for activities starting on or after 2026. Cases where information that could not be known is used for the prediction is deemed "leakage". Preventing leakage was an important factor in the methodology. Leakage has been a significant concern in other published works where forecasting with language models has been implemented, and in multiple cases has caused invalidation of forecasting results (Paleka et al., 2025). While ex-post evaluations may provide sufficient information to describe the activity, relying on language models to completely remove information about the eventual outcome could introduce information leakage from past the start date, and was a motivation behind using IATI, which explicitly separates documents into pre- and post-activity, as opposed to OpenAlex and 3ie ex-post activity documents.

Data Extraction LLM Model Selection Several different language models are used for data extraction pipeline, selected according to the following criteria:

- *gpt-3.5-turbo* is used when the task benefits from an earlier model training cutoff date, eliminating the possibility of temporal information leakage for events occurring after September 2021, in this case for used grading information documents. However, this legacy model is more expensive, despite the model having a weaker reasoning ability.
- *gemini-2.5-flash* was primarily used for its low cost, high speed, and ability to process pdf documents accurately
- *gemini-2.5-flash-lite* was used for high volume requests which are otherwise unaffordable, where the task complexity is also low
- *gemini-2.5-pro* is a more expensive model, used when many pages of PDF context and reasoning is required, with relatively few API calls

2.2 Data Sources

The methods of this thesis required a suitable data source which had all of the following: a sufficient volume of completed environmental and sustainability activities; paired pre- and post-activity documents to enable forecasting without future information leakage (see Section 2.1); machine-readable text extractable from database records; reliable temporal metadata; and the prospect of ongoing additions to allow future replication.

Several sources were evaluated against these criteria but ultimately not used:

- **The OpenAlex publication repository of peer-reviewed evaluation documents and abstracts** (Priem, Piwowar, and Orr, 2022)
  - Although many millions of evaluations are available on OpenAlex, it proved difficult to reliably identify and de-duplicate published evaluations of environmental interventions. Furthermore, OpenAlex provides only ex-post documents with no paired pre-activity description, making leakage prevention substantially harder.
- **The AidData database of international aid activities** (Tierney et al., 2011)
  - The AidData database has double-coded data entry. While double-coded data entry introduces less error than the LLM and regex data extraction techniques, the delay in hand-coding the ratings and results leads to relatively few recent environmental activities, and the lack of document links in AidData meant each activity record had far less extractable information. The overlap between AidData and the filtered IATI training set was less than 30% of IATI training set records. As AidData contained insufficient data for reliable forecasting on its own, and there was low overlap, merging IATI with AidData or enriching IATI with AidData was not attempted.
- **The 3ie development database of ex-post evaluations** (*3ie Development Evidence Portal | 3ie* 2025)
  - 3ie had fewer than 300 records for environmental topics, far lower than IATI, and provides only ex-post documents with no paired pre-activity description.
- **The IDEAL database of ex-post evaluations** (KfW Development Bank, 2025)
  - The IDEAL database had fewer than 300 records for environmental topics, far lower than IATI.

All activities for the thesis therefore come from the IATI database (*IATI Dashboard – IATI Activities* 2025). IATI met all selection criteria: it contains a large number of

activities with substantial textual information extractable from database records, and new records are continuously added, making future replication straightforward. Crucially, IATI explicitly separates pre- and post-activity documents by design, which makes preventing temporal information leakage tractable (see Section 2.1). Records contain reliable "actual start" and "actual end" dates, and commonly also report planned start and end dates. Most records contain an overall evaluation rating on a six-point scale within the linked evaluation PDFs. Activity records also reliably mark the reporting organization. The records used for this thesis have several activity information documents uploaded near the beginning of the activity, and several years later, at least one ex-post evaluation of the activity is uploaded as well, or results of key quantitative outcomes are sometimes directly recorded in the IATI database. The status of the activity is also always reported in the database, including if it is in the planning or completion/finalization stages, which is helpful information for forecasting.

A disadvantage of the IATI database is that it is sometimes inconsistent between reporting organizations as to how the data are filled in, and the format of documents was not always PDF, requiring conversion scripts from other formats to PDF. Many download links were also not functioning. Others required custom web-scraping scripts to extract project documents in PDF format from the original hosting websites. Planned start or end dates were often missing, leading to frequent exclusion of otherwise informative projects. Furthermore, approximately 30% of IATI activities do not have an activity category code, leading to a further exclusion of environmental or sustainability related activities that otherwise could have been included.

Country-level covariates were added from three standard sources: the World Bank governance indicators (Kaufmann and Kraay, 2024), the World Bank country policy and institutional assessment scores (World Bank, Operations Policy and Country Services, 2024), and GDP per capita (World Bank, 2024). These were included as supplementary features for the statistical forecasting models and were sometimes injected in the LLM forecasting prompts.

2.3 Data Filtering

The IATI dataset was progressively filtered from approximately 800,000 activities to 3,225 completed environment- and sustainability-related projects with both baseline and evaluation documents suitable for forecasting outcomes without future information leakage. Activities were first selected by sector codes (See Figure 1) and manual topic review, then constrained to those with downloadable documents, valid temporal ordering (baseline pdf uploaded at latest the end of first quarter of implementation period, evaluation at earliest last quarter), and sufficient metadata reliability. Additional restrictions ensured comparable rating scales and adequate sample sizes across four reporting organizations

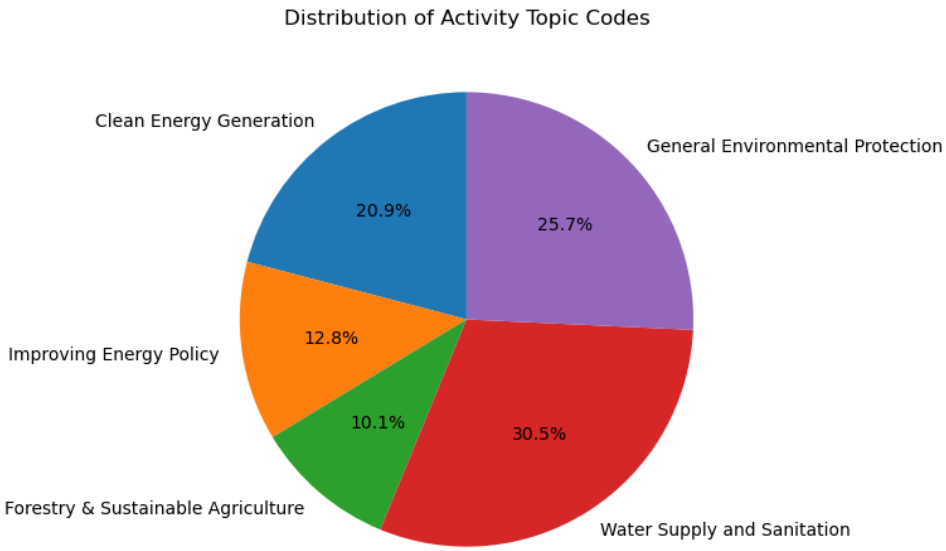

Figure 1: The split of topics analyzed from the dataset.

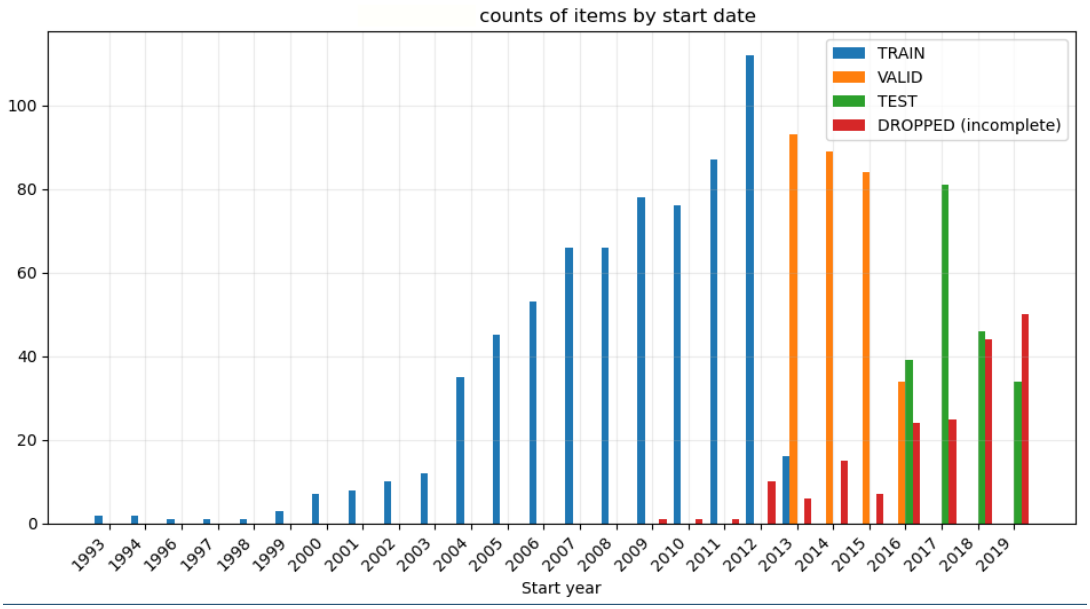

Figure 2: The activities included for forecasting ratings, with the splits by count year. Incomplete activities, shown in red, were not used for prediction. Activity ids used for outcome prediction were given the same temporal boundaries.

(World Bank, BMZ/KfW/GIZ, ADB, and UK FCDO, see Figure 3). Documents were converted to PDF and dated primarily using PDF metadata, which was validated against document content, yielding a median 22-day discrepancy. To prevent leakage from early progress reporting or misdated files, documents were screened by manual inspection, LLM-assisted checks, and automated keyword searches, resulting in the exclusion of activities showing substantive outcome progress. Further filtering removed purely legal or contractual documents from baseline and evaluation documents removed activities that were not marked as completed in IATI, and removed activities with no overall success evaluation ratings identified in the evaluation documents, ultimately producing a set of 1,181 activities which were used for the remaining analysis.

In order to accurately extract overall success ratings for each activity and useful textual information about the project for forecasting, each PDF document was processed using the following data processing pipeline:

2.4 Data Processing

All document pages had their rotation detected, and were rotated to vertical before processing via the Gemini API. Documents with ".odt", ".doc", or ".docx" extensions were converted to PDFs with a custom script. The pages when converted to PDFs were counted and zero-page documents were excluded.

Documents were then ranked and categorized, and structured information from the PDFs was systematically extracted via *gemini-2.5-flash*.

2.4.1 Ranking PDF Documents by Usefulness

Documents were first sorted by their pdf metadata date as pre-activity and post-activity. Pre-activity documents were created at the latest $\frac{1}{4}$ of the way through the activity and post-activity documents were created at earliest $\frac{3}{4}$ of the way through, although if a date was missing, standard titles just as 'Project Information Document' or 'Ex-Post Evaluation' were used to sort the documents into pre- and post-activity. Also, the documents were given a preliminary ranking (without an LLM query). Pre-activity documents that were closest to the activity start, and the latest post-activity documents were prioritized. Documents in a 3-20 page length were also prioritized above very short or very long documents.

Documents were then given a final ranking with *gemini-2.5-flash*, ranking from most to least useful for either forecasting the outcome for pre-activity documents or evaluating the results for post-activity documents. Each activity was processed in two separate API calls, one *gemini-2.5-flash* query for pre-activity documents and a similar *gemini-*

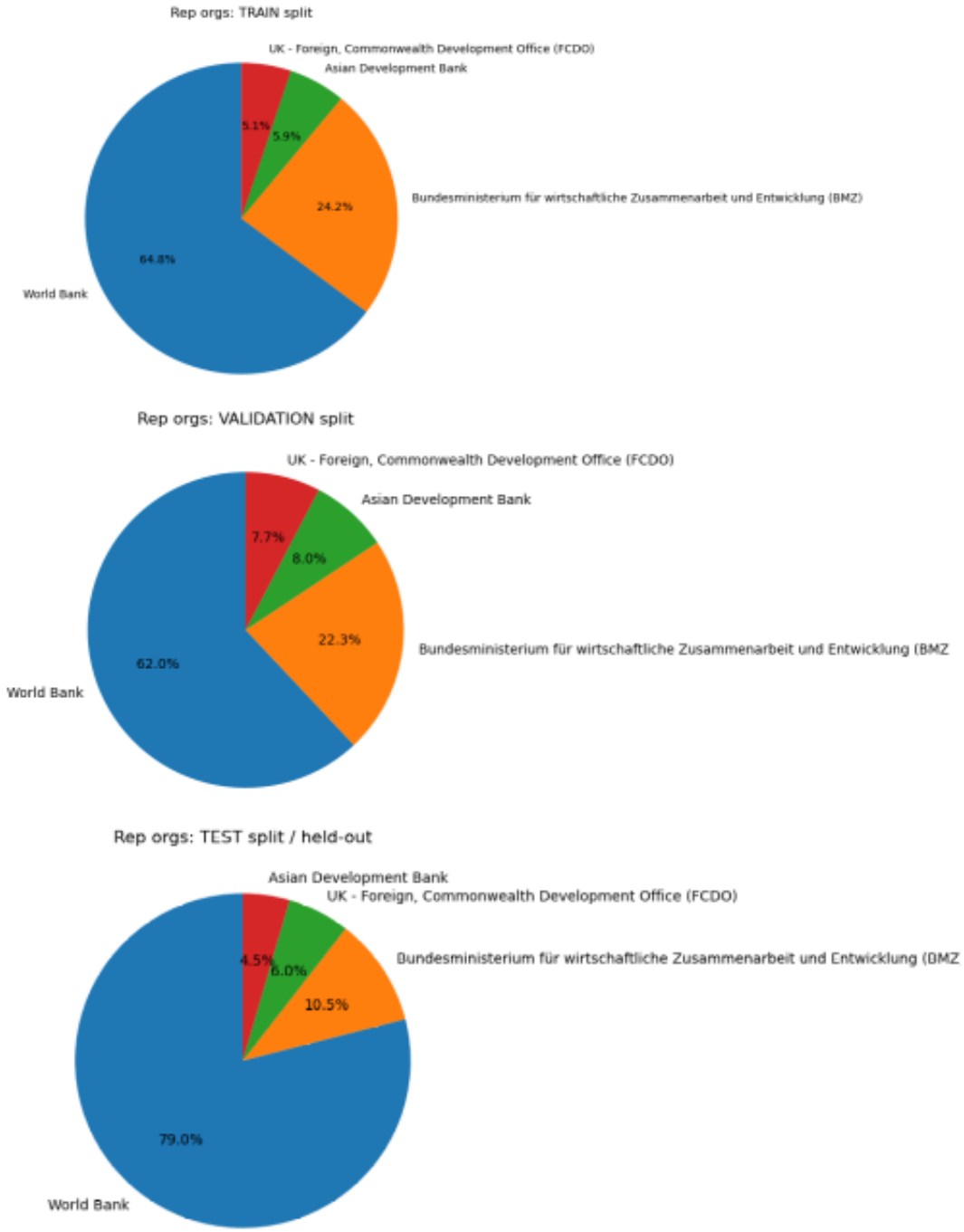

Figure 3: The breakdown of reporting orgs in the dataset which were used for training, validation and testing. The validation and test set contain differing ratios of reporting organizations between activities, making held-out set prediction more challenging.

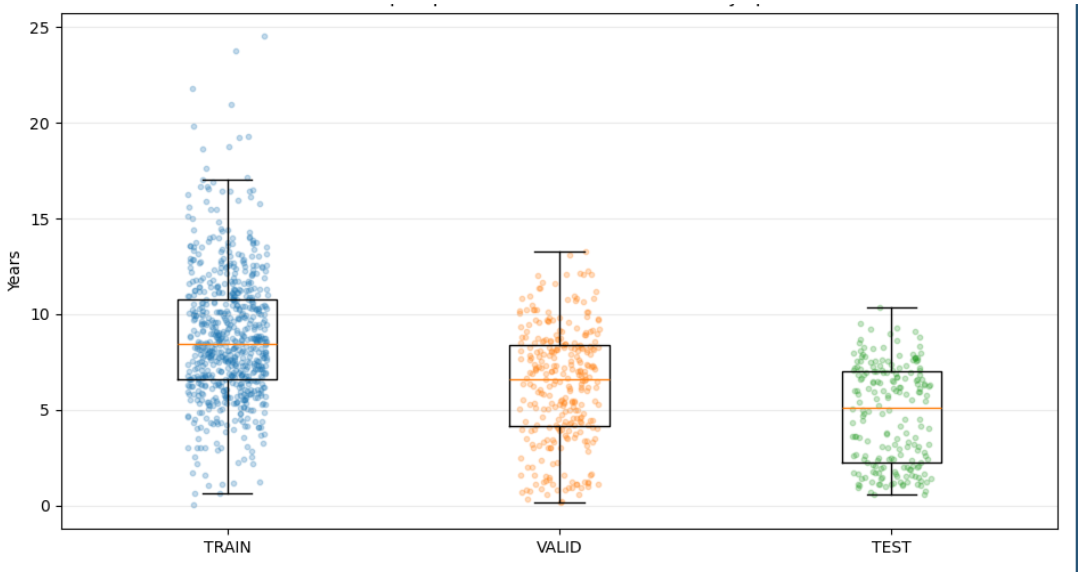

Figure 4: The durations of activities per split. More recently starting activities tend to be shorter, as they have not yet had time to complete and be evaluated. The out-of-distribution nature of validation and test sets increase the challenge of the model performing well on the held-out set.

2.5-flash query for post-activity documents. In each call, the first three pages of every candidate document were uploaded alongside a single prompt containing the activity title, description, official date range, and per-document metadata (title, language, IATI document category codes, page count, and PDF metadata date). A rubric specifying the information an ideal document would contain was included to anchor the grading scale. The model was instructed to return a single structured JSON response ranking all candidate documents from most to least relevant, assigning each a letter grade (A+ to F) and an include/exclude flag. Letter grades were selected in accordance with standard grading examples from Google Gemini documentation. Structured output was enforced via a JSON schema so that every document received exactly one rank and grade in a single response. Documents that were duplicates in a non-English language were excluded if the equivalent was available in English. For outcomes, if there were multiple progress reports, all the earlier ones were excluded and only the latest were kept in the rankings. After ranking, 2,312 documents had sufficiently informative activity information and activity evaluation documents.

2.4.2 Categorizing Pages within PDF Documents

The highest ranking documents then had each of their pages categorized with what information was present on them. Manual inspection determined this was more effective than semantic embedding search for long documents. To process the pages, PDFs were split into 3-page chunks. Each 3-page chunk was sent in PDF form to *gemini-2.5-flash*. Several random documents were read and categorized by-hand to understand common

sections, and the resulting categories, which informed the automated categorization selection. The pages were categorized differently based on whether the document was a pre-activity or post-activity document. Categories for outcomes allowed retrieval based on whether final evaluation in quantitative or qualitative form were present on the page, deviations from plans or other types of outcomes were detailed, or if the pages were simply overviews of the activity. Specifically, the allowed categorizations were "condensed summary", "sub activities outlined", "detailed implementation plans", "broad objectives", "possible outcomes", "quantitative targets", "qualitative targets", "risks as word or numeric", "risks or dangers generally", "plans to address key risks", "positive indicators", "progress reports", "similar cases outcomes", "implementation context country", "contextual challenges", "financing details", "budget and legal", "who implements", "whether part of larger program", "partner identity or skill", "whether skin in the game", "other stakeholder engagement", or "activity monitoring details" for pre-activity document pages, and "expected outcomes", "deviation from plans", "preliminary results", "final outcomes", "delays or early completion", "over or under spending", "overview as was planned", or "unrelated to evaluation" for post-activity document pages.

Upon manual inspection, it was revealed that pages often pertained to more than one category such as a page which documents final outcomes while also documenting project delays. The two most relevant category choices among these were therefore assigned for each page in the documents. Two categories were assigned in order to prevent less relevant categories from polluting the page tags.

In order to exclude irrelevant pages, the pages were also given a second categorization type, for post-activity document pages as "glossary", "blank page", "table of contents", "outcome evaluation", "activity description", "references", or "other", and for pre-activity document pages the same categories were options, in addition to "core activities", "theory of change", "targets", "broader context", and "preliminary results". Only one category choice among these was possible per page as these categories were found to typically be mutually exclusive.

2.4.3 Structured Information Extraction

2.4.3.1 Extracting Ratings

Two separate methods were used to extract ratings. Inspection of randomly selected pages revealed that ratings often appeared on pages with a high rating for relevance to evaluation, where a threshold of approximately 7/10 separated matches from bad fits. Therefore, the first method sends each individual post-activity document page rated above 7/10 for relevance to evaluation, or with a "quantitative targets" categorization, to *gemini-2.5-flash* to extract any overall ratings, and a second script summarized the overall ratings into a single value for the document. However, this was often insufficient

to capture the overall ratings. Another "fallback" script involved a custom generated word search with approximately 500 different rephrasings of "overall rating", "final result", "synthesized score", etc, in English, and searched the PDFs directly for an exact match on those terms, prioritizing pages with one or more exact text matches of such terms. Otherwise, if such words could not be found, the earliest pages in the document which were not categorized as "blank page", "appendix", "glossary", "table of contents", "references", or "activity description" were included and *gemini-2.5-flash* was queried to extract the overall rating from the documents.

For BMZ/KfW/GIZ documents, pre-activity documents were extremely rare. For this reason, the evaluation document was treated as a pre-activity document for the purposes of forecasting activity success. Categorization for these evaluations also was via the pre-activity document method described above. When grading or summarizing the features of the evaluation document, *gemini-2.5-flash* was instructed to only describe what could have been known at the beginning of the activity, and to under no circumstances reveal the final outcome of the activity. Manual inspection of extracted information from these documents did not reveal any resulting temporal leakage. BMZ/KfW/GIZ overall ratings were similarly predictable to World Bank ratings on the held-out set, despite this information leakage possibility.

2.4.3.2 Interpreting Ratings

Ratings were reported both with the rating itself, and a maximum and minimum possible rating. The World Bank rating scale from 1 ("Highly Unsatisfactory") to 6 ("Highly Satisfactory") was used as the template rating, and other ratings were rescaled to match the 1 to 6 rating scale. Notably, BMZ/KfW/GIZ ratings were inverted to reach this scale. A "Satisfactory" score was considered equivalent to scores such as "successful", "On Track", or "met expectations". Scores listed as percentages or fractions were re-scaled to the 1 to 6 scale as well. In order to ensure ratings were fairly compared, only the top four most common organizations with ratings were included for training and validation of the forecasting system.

2.4.3.3 Extracting cost-effectiveness

In addition to extracting ratings, a similar approach was used with *gemini-2.5-flash* to extract quantitative cost-effectiveness-related outcomes.

Cost-effectiveness of international aid activities was defined as the ratio of measured quantitative outcomes aiming to improve the environment or sustainability to the expenditures used to achieve that outcome. Quantitative outcomes were extracted by language model prompts as described below (the numerator), and the majority of international aid activities in IATI directly publish the overall activity expenditure (the denominator).

Quantitative outcomes were extracted from all pages which were categorized as containing outcomes, and marked as containing quantitative information. Unlike with ratings, extraction was not limited to the top 4 reporting organizations, as reporting ratings is more susceptible to between-reporting-organization variation and gaming than the reportable quantitative outcomes of projects (Goldemberg, Jordan, and Kenyon, 2025).

Once these PDF pages were extracted, a combination of manual examination of the extracted outcomes and a frequency analysis of bigrams was used to identify outcome variables that could be compared between projects. For each common outcome category, a list of words and phrases was compiled that would commonly match reports of these outcome variables in the description, as well as appropriate units. Each script ensures implausible or non-numeric values are dropped. Domain-specific filters also reduced false positives (e.g., wastewater context for pollution loads, water context for connections, agriculture context for yields).

Comparable Outcome Categories

Using manual inspection and frequency statistics of common bigrams, a set of outcome categories was defined that recur across evaluations and are interpretable across projects. One challenge encountered was ensuring sufficient quantities of comparable outcomes between activities. In order to ensure statistically significant quantities of activities were required for training, only outcomes categories were used that remained after ensuring each contained at least 50 activities in the training set. This threshold was selected to ensure reliable performance on the validation set, as below 50 points the model became unstable, while a higher restriction left too few points to evaluate the final prediction. The final set included:

- **Benefit/cost ratios (B/C):** benefit-cost ratio outcomes. (train: 62, val: 18, test: 15, total: 77)
- **Rates of return:** economic rate of return (ERR/EIRR) and financial rate of return (FRR/FIRR), in percent. (ERR — train: 396, val: 103, test: 43, total: 440; FRR — train: 190, val: 43, test: 15, total: 205)
- **Emissions reductions:** CO_2 or CO_2e reductions in tonnes. (train: 61, val: 23, test: 10, total: 72)
- **Water and sanitation connections:** counts of service connections, either new or repaired. (train: 96, val: 25, test: 9, total: 105)
- **Energy outcomes:** installed generation capacity, in MW (or occasionally GWh where the source reported capacity in energy units). (train: 64, val: 20, test: 12, total: 79)

Although pollution load removed, forest indicators, trees planted, air quality (PM2.5), clean cooking stove distribution, and agricultural yield improvement were parsed, none of these contributed sufficient counts to be useful for further outcome forecasting.

Multiple extracted values per category were aggregated by compiling all extracted outcomes in each outcome category for each activity and including them in an aggregation prompt for *gemini-2.5-flash* for aggregation into a final baseline, target, and outcome value. Where multiple extracted values conflicted, overlapped, or reflected different levels of aggregation (e.g., component-level versus project-level figures), the model was instructed to adjudicate between them and select or combine values to reflect a single coherent project-level outcome. The model was instructed to distinguish duplicates from additive components, prioritize project-level totals over subcomponent figures, aggregate only non-overlapping quantities, and return a null response when no coherent representative value could be determined.

Outcome aggregation and cost-effectiveness computation

Once outcomes were extracted and aggregated to project level, cost-per-unit estimates were constructed. The total disbursement for each activity reported by IATI served as the cost denominator (in USD), with the exception of benefit-cost ratios, rates of return, and agricultural yield outcomes, which are inherently unit-free or already normalised.

Outcome-level funding splits are not published in the IATI database. To avoid implicitly treating multi-outcome activities as having multiple full budgets, a custom algorithm allocates total activity expenditure across the non-overlapping funded sub-activities and activity reports. Benefit/cost ratios and economic and financial rates of return are excluded from monetary allocation; all other outcome categories are eligible. To avoid double-counting when two indicators are alternative measurements of the same underlying result, closely related indicators are first grouped into shared conceptual buckets (e.g., protected area paired with area under management; different yield-increase measures; tree planting paired with reforested area).

Once the components are bucketed, each bucket receives an equal share of the activity's allocatable funding. Every component inside a bucket inherits that same share, so "alternative measures of the same thing" draw on one portion of the allocation rather than each taking a separate slice.

Carbon dioxide reductions are handled as a special case because they can act as a summary metric that overlaps with other mitigation outputs. If CO_2 reductions are reported without any closely linked mitigation outputs (such as improved stoves, added generation capacity, or trees planted), CO_2 reductions receive an equal share like any other bucket. If CO_2 reductions are reported alongside any of those linked outputs, they inherit the combined allocation already assigned to the CO_2-mitigating expenditures for that activity, preventing CO_2 from inflating allocated spending when it is a co-reported consequence of other outcomes.

Empirical inspection found that most cost-per-unit outcomes are approximately log-normally distributed. The $\log_{10}$ transform was therefore applied to all outcome categories

except benefit-cost ratios and rates of return before modelling.

The cost-effectiveness aggregate Z-score is standardised using training-set mean and standard deviation (population stddev, ddof=0) applied to all splits. The Random Forest used for cost-effectiveness forecasting is trained with 3× sample weight on the most recent 20% of training observations, as this improved performance on the validation set (and later on the test set). A ridge regression on the residuals with $\alpha = 50$, as used for outcome tags and overall rating prediction, was also applied due to its performance improvement on the validation set.

2.4.3.4 Extracting Outcome Tags

All evaluations were summarized, common outcome tags were assessed from a random sample of these summaries, and then the value of each true/false tag was determined for all outcome summaries.

Extracting Outcome Summaries

For each activity, up to ten post-activity document pages were selected for outcome summarization, prioritising pages categorised as containing deviations from plans, delays or early completion, and over- or under-spending, with surrounding pages included when insufficient high-relevance pages were available. If an activity had no categorised pages meeting the threshold, a fallback retrieved the highest-scoring uncategorised outcome pages directly from the document. These pages, together with the activity title, planned date range, scope, and an optional pre-activity description (where available), were submitted in a single *gemini-2.5-flash* prompt instructing the model to return 3–10 paragraphs summarising what occurred as a result of the activity. The model was instructed to report both successes and shortfalls and any quantitative outcomes relating to the activity's core aims while explicitly withholding any overall evaluation rating. The resulting outcome summaries were used as the input for outcome tag classification.

Discovery of Common Tags

The tag taxonomy was developed inductively from the evaluation corpus itself rather than imposed from an external framework. A random sample of 250 outcome summaries from completed activities was drawn. This sample was divided into batches of 10 outcome summaries each. Each batch was submitted to *gemini-2.5-flash* with a prompt requesting identification of all recurring outcome types present across the summaries in the batch. A set of seed examples determined from manual inspection of random activity summaries was provided to anchor the initial vocabulary (e.g., "implementation delays," "high beneficiary satisfaction," "project was restructured," "capacity building training was delivered"), and the model was instructed to generate additional tags for any distinct outcome type appearing in more than one summary. All 25 batches were then deduplicated and merged via manual inspection, resulting in 23 outcome tags.

Unsigned and Signed Tags

The 23 tags were divided into two conceptually distinct types based on whether their presence implies a value judgement.

Of the 23 tags, 11 were interpreted as not having occurred if not mentioned. These are applicable to any project and would be very likely to be mentioned in the summary if it occurred. For example, whether the project was delayed. The remaining 12 tags were not recorded as false if missing. Instead, information as to whether they occur was recorded only if explicitly mentioned, such as having successfully completing or failing to complete infrastructure, or whether livelihoods were significantly achieved by the project. For unsigned tags, the LLM is asked whether the phenomenon was clearly described as occurring, and the response is a single boolean per tag.

The 23 tags are organized across four broad categories: finance and budget outcomes, activity rescoping outcomes, process and implementation challenges, and target achievement outcomes.

Applying Tags at Scale

The finalized set of tags were extracted for all activities with overall activity ratings within the four reporting organizations using two separate LLM passes over each activity's post-activity outcome summary. In the first pass, all 12 unsigned tags were evaluated together in a single prompt per activity, with the model asked to return a JSON object mapping using structured JSON output enforced via a schema.

2.5 Baseline Methods

Always predict the most common rating for the reporting organization

This baseline technique provides a sanity check that more sophisticated methods are worthwhile. If this baseline outperforms more sophisticated models, it indicates that much of the methodology is overfitting on noise in the data.

Ridge regression with reporting organization and risk score

In order to assess how well calibrated overall risks are in activity documents, a ridge regression model was trained given both the LLM grade for the degree of risk assessed at the beginning of the project, as well as the dummy variable representing which organization was doing the rating. While by construction the full forecasting system also includes these features and thus with sufficient regularization must perform better than this baseline, it does provide some insight into how well we can infer the eventual rating from the risks alone in activity documents at approval. This score was only calculated for activities where the LLM was able to successfully extract the grade for how risky the project is from

the pre-activity documents. This method suffers from potential inconsistency in LLMs extracting the overall risk from pre-activity documents.

gemini-2.5-flash summary of risks

A separate baseline used only for assessing the forecast grades of models, is submitting a summary of the risks identified from activity documents to the *gemini-2.5-flash-lite* grading LLM, and comparing these to proper LLM forecasts. This is a separate measure of the informativeness of risks identified in activity documents, and provides a sanity check that the LLM is not simply summarizing what the authors of activity of appraisal documents already highlight as risks. However, the lack of assertive claims about what will happen in a positive sense hinders this baseline metric relative to graded forecasts.

deepseek-V3.2 provided only the activity summary and statistics

Several LLM methods are introduced, closely informed by prior state-of-the-art methods in judgemental forecasting literature (Halawi et al., 2024) (Lee et al., 2025). In order to justify these methods, comparison is made to a simple baseline where all provided to the model is the activity ID, the activity title, an activity summary from *gemini-2.5-flash* of the 10 most important pages from the activity documents for forecasting the outcome, and the statistical prevalence of each outcome rating, and ask it directly to forecast the evaluation rating.

Constant base-rate prediction for outcome tags

For binary outcome tags, the no-skill baseline predicts the training-set positive rate $\bar{p}_{\text{train}}$ for every activity, regardless of its features. This constant forecast is the natural reference for the Brier skill score: a model that achieves BSS > 0 beats this baseline, while BSS $= 0$ indicates the model adds no probabilistic skill over simply forecasting the base rate. The same constant baseline also sets the accuracy of a no-feature classifier: a classifier that ignores all inputs and always predicts the majority class achieves accuracy $\max(\bar{p}_{\text{train}},\, 1 - \bar{p}_{\text{train}})$. The accuracy ratio reported for each tag (Table 3) is therefore the model accuracy divided by this constant-baseline accuracy, with values above 1 indicating improvement.

Ridge regression trained with non-LLM categories

In order to justify the addition of non-LLM categories, the baseline statistical categories used in prior literature are used and a ridge regression model is trained on the outputs. All features included improve performance on the validation set for the final forecasting system, reducing performance when removed. Sources are cited which have found meaningful predictive effects or correlations between activity success or activity outcomes and the features listed.

The non-LLM features used were:

- planned activity duration (Goldemberg, Jordan, and Kenyon, 2025)
- planned total disbursement (consultation with BMZ officers), (Bulman, Kolkma, and Kraay, 2017) (Goldemberg, Jordan, and Kenyon, 2025) (H. L. Ashton et al., 2021) (Eilers et al., 2025)
- derived features: log(planned total disbursement), planned total disbursement per year
- whether the activity is primarily loan or grant-based (Bulman, Kolkma, and Kraay, 2017)
- the one-hot encoded reporting organization (Goldemberg, Jordan, and Kenyon, 2025)
- the Country Policy and Institutional Assessment (CPIA) score from the World Bank for that country (Bulman, Kolkma, and Kraay, 2017)
- the World Governance Indicators scores (Kaufmann and Kraay, 2024) for control of corruption, government effectiveness, political stability, regulatory quality, and rule of law, as well as the mean of these values when none are median imputed (consultation with BMZ officers), (Goldemberg, Jordan, and Kenyon, 2025) (Ndikumana and Pickbourn, 2017)
- the scope of the activity on a scale from 1-7, ranging from local to global (Vivalt, 2020)
- the log(GDP/capita) of the countries where the activity takes place weighted by the percentage of the activity performed in each country. (Goldemberg, Jordan, and Kenyon, 2025) (Bulman, Kolkma, and Kraay, 2017)
- A dummy variable for each region as defined by the World Bank (AFE, Eastern and Southern Africa; AFW, Western and Central Africa; EAP, East Asia and Pacific; ECA, Europe and Central Asia; LAC, Latin America and the Caribbean; MENA, Middle East and North Africa; SAS, South Asia) (Goldemberg, Jordan, and Kenyon, 2025) (Vivalt, 2020)
- missingness counts for important features with missingness

The activity start date was not used as a feature, as there was no linear pattern with regards to overall activity success over time in the training data, and adding it tended to increase overfitting and reduce out-of-time generalization. As opposed to some results in prior literature (Vivalt, 2020) (H. L. Ashton et al., 2021), whether the implementer was a government, NGO, or another category was also not found to have any improvement in model performance and was excluded.

2.6 Experimental Methods

This thesis introduces several novel techniques which have not been assessed by the literature. These are deemed "Experimental Techniques", and are rigorously assessed as

to whether they legitimately improve forecasting system skill.

2.6.1 Statistical Forecasting Methods

Nearest Neighbor (Vector Similarity)

A similarity test was first constructed using features including countries of the activity, GDP per capita as described previously, the scope of the activity, and the implementing and funding organization ID. This similarity test was found, however, to significantly underperform compared to the semantic similarity of the *gemini-2.5-flash*-generated summary of the activity documents, so this similarity method was used instead. Similarity was first weighted proportional to its embedding semantic similarity score, and a cutoff was tested for averaging 1, 3, 7, 10, 15, and 20 nearest neighbors using the Gemini embeddings model *gemini-embedding-001*. 15 nearest neighbors was found to be the highest-performing number of neighbors using this method. The weighted average of the nearest neighbor ratings is used to predict the overall activity score.

Although the nearest neighbor method was used to collect examples for the LLM prompt, it was found that simply taking the weighted mean of 15 similar ratings performed worse than the "most common rating" method and thus was not used as an experimental method to directly forecast ratings.

Per-Organization Mode Differencing

All statistical models (ridge regression, Random Forest, Extra Trees, and XGBoost) are trained not on the raw ratings y_i but on per-organization residuals $\tilde{y}_i = y_i - m_o$, where m_o is the mode of the training ratings for reporting organization o. The per-organization mode m_o is computed from the training set only and is used as a fixed offset for all activities from that organization. Predictions from each model are therefore organization-relative residuals; the final predicted rating is recovered by adding back m_o:

$$\hat{y}_i = \hat{\tilde{y}}_i + m_o.$$

This decomposition removes the dominant between-organization rating shift before model training, allowing models to focus on within-organization variation rather than fitting the coarser organization-level mean. The per-organization mode also serves as the mode baseline described in Section 2.5.

Random Forest

The Random Forest (RF) method is a statistical algorithm which constructs an ensemble of decision trees which would produce the correct output on the training data, and averages those decision trees. The averaging nature of the RF algorithm reduces overfitting on the training data. The algorithm is inherently "regularized", penalizing an overly complex

decision tree. The decision trees split based on value ranges of the features. By reducing the depth of the trees (the number of decision points where the decision tree splits), we can reduce the memorization of the training data from the trees, and improve generalization of the model. Each decision also only considers a random fraction of the features, allowing each tree to be more independent of each other and improving generalization further. The bootstrap method is also used to train trees, encouraging tree independence.

Optimal parameters for the overall rating RF+ET were discovered via a combination of sweeps over parameter space and LLM-generated suggestions iterating on the identified best parameters. Relative to the default `RandomForestRegressor` configuration (e.g., `n_estimators`=100, `max_depth`=`None`, `min_samples_split`=2, `min_samples_leaf`=1, `max_features`=1.0, `bootstrap`=`True`, `ccp_alpha`=0.0), the best-identified specification deliberately constrains model capacity in ways that typically reduce overfitting. Trees are explicitly depth-limited (`max_depth`=14 rather than unbounded) and splits are only permitted when nodes contain substantially more data (`min_samples_split`=20 and `min_samples_leaf`=20), which smooths forecasts by limiting fine-grained partitioning of the feature space. In addition, using a smaller feature subset at each split (`max_features`=0.488) increases tree diversity and reduces variance relative to the default that considers all features. The model also uses row subsampling (`max_samples`=0.86), further reducing variance by injecting additional randomness into each tree's training set. Compared to defaults, these choices accept slightly less precise fits on the training data in exchange for predictions that are more stable across unseen examples.

Extra Trees The Extra Trees (ET) method, short for Extremely Randomized Trees, is a variant of the Random Forest algorithm that constructs an ensemble of decision trees with an additional layer of randomization. While Random Forest selects the optimal split point for each randomly chosen feature subset, Extra Trees goes further by also randomizing the split thresholds, drawing cut points uniformly at random from each feature's value range and selecting the best among those random candidates. This additional randomization further decorrelates the individual trees in the ensemble, reducing overfitting at the cost of a slightly less precise fit to the data. Unlike Random Forest, Extra Trees trains each tree on the full training dataset rather than a bootstrap sample, which when ensembled with Random Forest is beneficial as it ensures the two models derive their diversity from different sources: one from varying the data each tree sees (bootstrap), and the other from randomizing split logic (ET). This makes their errors more complementary and the ensemble average more effective. As with Random Forest, predictions are made by averaging the outputs of all trees in the ensemble.

XGBoost

The XGBoost (Extreme Gradient Boosting) method is a statistical algorithm which constructs an ensemble of decision trees sequentially, where each subsequent tree attempts

to correct the errors made by the previous trees. Unlike the RF which trains trees independently and averages their predictions, XGBoost builds trees iteratively, with each new tree focusing on the residual errors of the ensemble.

The algorithm incorporates both L1 and L2 regularization terms to penalize model complexity and prevent overfitting on the training data. The decision trees split based on value ranges of the features, using a splitting criterion that accounts for the gradient and hessian of the loss function.

LLM-generated scores

Seven additional features were extracted using *gpt-3.5-turbo*-generated evaluation on a score from 0 to 100 in order to enrich the activity with information from the project document PDFs from the start of the activity. GPT-3.5-turbo was used in order to reduce leakage of information after the September 2021 model cutoff date. Inspection of grades did not reveal significant usage of knowledge that would not be available at the beginning of the activity. The majority of activities in the test set ended after the model cutoff date. These seven features were based on important aspects revealed by the literature review and consultation. One feature not introduced by others was added (the ease of targeted outcomes). Unlike technical complexity, which captures only one dimension of difficulty, or overall risk, which aggregates financial, legal, and regulatory threat sources, ease of targeted outcomes is distinct. It specifically reflects whether the activity's intended outputs are inherently measurable and achievable given typical implementation constraints—a factor theorized to independently contribute to evaluation success.

The features added for this method included the seven features and one interaction term

- *gpt-3.5-turbo*-generated evaluation on a score from 0 to 100 of:
  - how well financed the activity is (consultation with BMZ officers), (Bulman, Kolkma, and Kraay, 2017) (Goldemberg, Jordan, and Kenyon, 2025) (Eilers et al., 2025)
  - the activity integratedness within the broader activity ecosystem (consultation with BMZ officers), (Vivalt, 2020)
  - the expected implementer performance including ownership (consultation with BMZ officers), (Bulman, Kolkma, and Kraay, 2017) (Goldemberg, Jordan, and Kenyon, 2025) (H. L. Ashton et al., 2021)
  - the degree of contextual challenge (consultation with BMZ officers)
  - the overall risk level (consultation with BMZ officers), (Eilers et al., 2025)
  - the activity's overall technical complexity (Eilers et al., 2025)
  - the ease of targeted outcomes
- Additional interaction term added

– expenditure times complexity

Language Model Embeddings

The underlying information contained in text can be mapped onto a high-dimensional vector of numbers. This process is called "embedding" and is performed with language models. In this case the textual input to the embedding is the LLM-generated "targets" field, which summarizes the objectives of what the activity aims to accomplish. This follows (Goldemberg, Jordan, and Kenyon, 2025) in extracting the World Bank Project Development Objectives (PDO), as they are found to be very similar to the LLM-generated outputs. PDO extraction was also attempted using regex methods, but the results were found to be noisy on matching, especially on projects not from the World Bank, and did not improve prediction performance as much as embeddings on the LLM-generated targets. The LLM-extracted targets text was first normalized into a stable canonical form (removing formatting artifacts, unescaping, splitting on separators, dropping "NO RESPONSE" tails, and deduplicating near-identical chunks). The cleaned targets text for each activity was then embedded with the *gemini-embedding-001* model, yielding a single high-dimensional vector per activity that semantically encodes activity objectives.

Following (Goldemberg, Jordan, and Kenyon, 2025), target embeddings are compressed using a two-stage dimensionality reduction pipeline. This works via Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) (Healy and McInnes, 2024). PCA projects embeddings into a few directions that preserve most of their variation by finding orthogonal directions along which the data varies most, then projecting the data onto those directions. UMAP builds a graph of nearest neighbors in high-dimensional space, then optimizes a low-dimensional embedding to preserve those relationships. PCA was first used to reduce the embedding vectors to 50 dimensions and then UMAP was fitted on the PCA outputs to produce 2D and 3D coordinate maps. The 3D embeddings (umap_x, umap_y, umap_z) were found to perform better on prediction on the validation set for forecasting activity ratings than 2D or 4D, and thus 3 dimensions were chosen. This preserves enough local topology that activities with similar targets remain near each other in the compressed space. Because the evaluation uses an out-of-time split, PCA and UMAP are fitted only on activities starting before the relevant evaluation cutoff: training-period activities only when evaluating on the validation set, and train+validation activities when evaluating on the test set. Held-out activities are never included in the manifold fit; instead their PCA-compressed embeddings are projected into the pre-existing UMAP space via the learned `transform()` function. This prevents any future-activity information from influencing the coordinate system, at the cost of somewhat noisier projections for activities lying outside the training distribution. The same cutoff logic applies to the sector and country centroids used for the distance features described below. Visual inspection also revealed that sectors with similar 2D vectors cluster around the activity environmental category, replicating the finding in (Goldemberg,

Jordan, and Kenyon, 2025).

By counting the word occurrences between features higher or lower on the UMAP x,y, and z axes, it is possible to investigate what aspects are reported by the embeddings, and the common sectors which they were categorized in. Broadly, low x values correspond to forestry, agriculture, water, and management related terms while high x correspond to energy sector and financing related terms. Low y correspond to similar terms as high x, both mostly in the energy sector. High y terms related to biodiversity, wastewater, and conservation. The z axis seems to relate more to urban vs rural: Low z corresponds to wildlife and undp related terms as well as "rural", while high z corresponds to sanitation and wastewater, as well as "urban" and "city". The z axis may also correlate slightly with contextualization, where the more "cookie cutter" objectives with regards to the sector in question are typically correlated with low z values. Lastly, the z axis also has a weak negative correlations with the number of countries: low z values tend to occur in fewer countries.

This method adds 4 additional features to the model:

- umap3_x: First UMAP coordinate capturing a spectrum from forestry, agriculture, and water management objectives (low values) to energy-sector and financing-oriented objectives (high values).
- umap3_y: Second UMAP coordinate separating energy-focused activities (low values) from biodiversity, wastewater, and conservation-oriented activities (high values).
- umap3_z: Third UMAP coordinate primarily reflecting an urban–rural axis, with lower values associated with rural and wildlife objectives and the UNDP, and higher values associated with urban sanitation and wastewater.
- a binary indicator for whether the umap embedding is missing (missing features are always imputed using the median values in the training dataset)

Language Model Embedding Contextualization

Beyond the compressed UMAP coordinates, two scalar distance features are derived from the full-dimensional target embeddings. All embeddings and centroids are L2-normalized, so the Euclidean distance computed is equivalent to cosine distance. For each activity, *sector distance* is the distance between the activity's L2-normalized target embedding and the L2-normalized centroid of all target embeddings belonging to the same DAC5 sector and decade group. *Country distance* is the distance between the activity's embedding and the L2-normalized weighted sum of the recipient countries' embedding centroids, where the weights are the fraction of the activity carried out in each country, again stratified by decade; if no country data are available the decade-level centroid is used as a fallback.

Both centroids are fitted on activities up to the relevant evaluation cutoff, falling back to all activities for groups with fewer than a threshold number of training examples. These two features therefore capture how semantically typical an activity's stated objectives are relative to its sector peers, and relative to the portfolio of activities in its recipient countries, respectively. Both were found to be informative on the validation set and are included in the full feature set.

This approach adapts and extends the methodology of (Goldemberg, Jordan, and Kenyon, 2025), who use Sentence-BERT embeddings of Project Development Objectives and project titles to construct sector- and country-level distance measures for World Bank projects, and show these embeddings correlate with quantitative country-level sector outcomes. The key differences here are that the present work uses *gemini-embedding-001* rather than Sentence-BERT, trains on LLM-generated "targets" text rather than raw PDO strings (enabling coverage beyond World Bank activities), and computes distances at the activity level for individual success prediction rather than in aggregate for country-level outcome analysis. For fuller methodological details see (Goldemberg, Jordan, and Kenyon, 2025).

This method adds 2 additional features to the model:

- sector_distance: Euclidean distance (equivalent to cosine distance, as all vectors are L2-normalized) between the activity's target embedding and the centroid of target embeddings for activities in the same DAC5 sector and decade. Low values indicate the activity's objectives are typical for its sector; high values indicate an unusual focus.
- country_distance: Euclidean distance between the activity's target embedding and the L2-normalized weighted sum of per-country centroids for the recipient countries, weighted by the fraction of the activity carried out in each country (decade-stratified). Low values indicate the activity's objectives closely resemble the typical portfolio of its recipient countries; high values indicate objectives that are atypical for those country contexts.

Budget allocation for sector clustering

A key component of the efficacy of projects is how they allocate their funding. To do so, *gemini-2.5-flash* was queried to identify the breakdown of outcome-related funding, where available in the pre-activity documents. Funding breakdowns were found to be important in prior literature (Goldemberg, Jordan, and Kenyon, 2025). However, unlike in (Goldemberg, Jordan, and Kenyon, 2025), the measure of how much the funding allocations are split between different domains (the Herfindahl–Hirschman index (HHI)) was found not to improve performance on the validation set and was not added as a feature. The allocations of funding are required to approximately sum to the total expenditure of the project, with approximately 68% of projects having their budgets identified. If no

high-scoring finance or budget pages were identified in the categorization step, the first 10 pages of the highest-ranked pre-activity document were used as a fallback.

In order to make the resulting dataset of budgets usable by the statistical model, the embeddings model was used to cluster the descriptions of the subsectors into 15 sector clusters (having tried 10, 15, and 20 clusters, 15 was found to provide the optimal performance on the validation set). The allocations within each cluster are summed and reported as a fraction of the total as a feature for the statistical model.

This adds the following 16 features to the statistical model:

- 15 features, one for each sector cluster expenditure percentage (Goldemberg, Jordan, and Kenyon, 2025)
- a binary missing indicator if there is no sector cluster information in the pdf. As for other features, the result is median imputed, meaning all features are set to 0 as that is the median for all sectors in the training data.

2.6.2 Outcome Tag Forecasting

For each outcome tag, an RF+ET ensemble classifier was trained using the same initial feature set and time-ordered split as the overall rating model (train (start date on or before February 6, 2013; validation: February 7, 2013 through June 6, 2016; test: after June 6, 2016 and after January 1, 2020, see Figure 2).

The model for the outcome tag was judged as ineligible for test-set prediction unless the Brier skill score was strictly positive and the within-group pairwise ranking probability exeeded 50%. Otherwise the chosen model was reverted to the constant baseline, indicating low confidence that the RF+ET ensemble would perform better than the validation set.

The forecasted tags were:

- **Funds Cancelled or Unutilized:** Project funds were cancelled, not disbursed, or significantly underutilized.
- **Funds Reallocated:** Project funds were reallocated across components or activities.
- **High Disbursement:** Project achieved high or full disbursement of funds.
- **Improved Financial Performance:** Improved revenues, profitability, financial management, or financial capacity of beneficiary institutions were significantly achieved.
- **Project Restructured:** Project was formally restructured, including changes to scope, components, or objectives.

- **Closing Date Extended:** Project closing date was extended beyond the original plan.
- **Targets Revised:** Project targets, indicators, or results framework were revised during implementation.
- **Policy and Regulatory Reforms:** Policy, regulatory, or legal frameworks were significantly and successfully developed, strengthened, or operationalized.
- **Targets Met or Exceeded:** Project targets were met as planned, or exceeded.
- **Over Budget:** Project had to exceed the original budget.
- **Capacity Building Delivered:** Capacity building, training, or institutional strengthening activities were significantly delivered.
- **High Beneficiary Satisfaction or Reach:** High beneficiary satisfaction, broad beneficiary reach, or high uptake was achieved.
- **Gender-Equitable Outcomes:** Outcomes affecting women and girls were equitable or positive.
- **Private Sector Engagement:** Success in attracting private sector investment or improving the business environment was achieved at or above targets.

The tags which were not predicted on the test set due to poor validation performance were as follows: Monitoring and Evaluation Challenges, Energy Sector Improvements, Improved Livelihoods, Improved Service Delivery, Infrastructure Completed, Activities Not Completed, Design or Appraisal Shortcomings, External Factors Affected Outcomes, Implementation Delays. Many of these had very few negative examples or a highly imbalanced true/false split, leading to the weak model performance.

Feature selection

An ablation experiment for all tags was conducted on the validation set. The RF+ET model uses 63 independent features, which may not be appropriate for all outcome tags. A series of experiments were conducted to assess which features should be used per-tag.

Features were grouped into llm grades, governance indicators, regions, budget, org structure, UMAP features, sector clusters, economic context, and missingness flags.

Features were also assessed on a per-tag basis. Use of all non-ablated features for all tags was found to reduce model performance, as some tags were unrelated to many features, and the imbalanced True/False divide and lack of tag data for some training activities meant that the effective training data are much smaller for some tags and thus are at a higher risk of overfitting.

A feature-selection sweep tested subsets of the top-5, top-10, top-30 features ranked by RF feature importance on the training-set of the full set of input features from the overall rating model. To select among these candidates for each tag without capitalising on a by-chance best result, a pairwise tournament across brier skill score, accuracy ratio, and within-group pairwise ranking skill was used.

Small losses on one metric (below 1.5 percentage points) were forgiven if a compensating gain on another metric was at least 20% of the loss. The candidate with more net wins was selected; ties were broken by Brier skill score.

Because this selection was made on the same validation set used to evaluate final performance, there was a risk that the feature reduction reflected overfitting to that split rather than a genuine regularisation benefit. To check this, the selected feature set for each tag was re-evaluated inside the three-fold temporal cross-validation on the training set alone, where the validation set played no role.

Ultimately, it was unclear whether per-tag aggressive feature reduction or a more permissive set of features would generalize better to the held-out set. Therefore, both the limited per-tag feature set and the full-feature set were used for the forecast on the held-out test set.

A group-level ablation additionally revealed that governance indicators (World Governance Indicators, CPIA score, and a governance × complexity interaction term) and missingness-flag features were introducing more noise than signal. These 18 features (7 governance, 11 missingness) were dropped from all tags, leaving 45 features available for per-tag selection. The drop was confirmed to improve scores on the k-fold temporal cross-validation on the training set. For each tag assigned a reduced feature subset, k-fold performance was at least as good as the full-feature alternative, indicating that the reduction reflects a genuine generalisation benefit.

RF+ExtraTrees ensemble

RF and ET both used classifier models rather than regressors as in the ratings training, as the method is intended for binary probabilistic prediction rather than a regression prediction. XGBoost, and LightGBM were both tested against the RF+ET model, but early experiments showed they performed significantly worse and use of other models was not continued.

In the case of binary tags, three options were evaluated: The `class_weight` parameter was set to always `none`, always `balanced`, and a tag-dependent scheme. The tag-dependent scheme selected a threshold where above 65% of training examples were positive, `class_weight=None` was used allowing the model to predict near the natural base rate, and below 65% `class_weight="balanced"` was used to prevent the ensemble from collapsing to the majority class.

Optimal parameters for the RF+ET classifier ensemble were identified by exploring 20 LLM-generated parameter sets, with the second set of 10 varied near the maximal score of the first set. These included variations in tree depth, leaf size, feature sampling width, row subsampling, and pruning strength, as well as between-RF and ET variations.

The tag-dependent class weight scheme was found to perform best across most tags on the validation set and was used for the final model. The best-performing configuration was: `min_samples_leaf`=5, `max_features="sqrt"`, `n_estimators`=500, `max_depth=None`, `min_samples_split`=2, `ccp_alpha`=0, with bootstrap `True` for the RF and `False` for the ET. XGBoost and LightGBM were both tested and performed worse on every tag assessed.

The RF+ET ensemble was otherwise trained identically to how ratings were trained.

Start-year trend correction

After training, predictions were corrected for a linear temporal trend in the residuals. A ridge regression (penalty $\alpha = 50$) was fitted on training data to predict the residual $r_i = y_i - \hat{p}_i$ from the activity start year:

$$\hat{r}_i := \gamma_0 + \gamma_1 \cdot \text{year}_i,$$

and the corrected probability was:

$$\hat{p}_i^{\text{corr}} := \text{clip}_{[0,1]}(\hat{p}_i + \hat{r}_i)\,.$$

The strong ridge penalty shrinks the slope toward zero to avoid overcorrecting on noisy year–residual relationships. This correction was adopted wherever it improved validation set performance. The correction multiplier was validated by sweeping values in $\{0, 0.1, 0.2, 0.5, 1.0, 1.5, 2.0\}$ and evaluating mean within-group pairwise ranking, Brier skill, and accuracy skill on the validation set across all corrected tags. A multiplier of 1.0 (full correction) achieved the best mean pairwise ranking, Brier skill, and accuracy, and was therefore retained. The correction was applied to the RF model alone before averaging with ET, as this was found to outperform correcting the ensemble average on the validation set. The correction was applied to all tags except `tag_monitoring_and_evaluation_challenges`, where the year trend was found to hurt validation performance.

Tag group averaging

Analysis of outcome tags in the training data revealed that certain groups of tags are strongly correlated. For example, tags relating to targets being met correlate with good financial performance. Two groupings were identified in the data which improved performance on the validation set when incorporated into predictions:

- Success Group: average of six success-related signed tags (targets met, beneficiary

satisfaction, private sector engagement, capacity building, policy reforms, and financial performance).

- Financing Group: disbursement tag, funds-cancelled tag (inverted).
- Rescoping Group: Project Restructured, Targets Revised, Closing Date Extended, Funds Reallocated. This group's RF+ET regressor is trained and its predicted score is used as an OLS predictor input when reconstructing per-tag probabilities for blended tags, but rescoping-group tags are not directly blended (the group influences other tags via OLS rather than being a direct mixture target).

For each group, an RF+ET regressor on the 45 features was trained to predict the group score from features. The per-tag probabilities were then reconstructed from predicted group scores via a OLS linear regression fitted on training data.

It was observed that the group average probability was typically more informative for low-data tags. For this the minority class count in training was used, which is the minimum between the number of true or false tags in the training data. 50 different blend selection options were sweeped across with various thresholds for the number of training points in the minority class to set as thresholds and various mix ratios of the group and tag average after the threshold.

The best performing blend on the validation set was a linear increase from 100 to 400 minority-class training points:

$$\hat{p}_i^{\text{final}} = w \cdot \hat{p}_i^{\text{model}} + (1 - w) \cdot \hat{p}_i^{\text{group}},$$

where $w = \text{clip}\left(\frac{m-100}{400-100}, 0, 1\right)$ and m is the minority class count. Below 100 training points the group average is used exclusively; above 400 the per-tag model prediction is used exclusively.

LLM correction

An LLM-based correction step was tested for tag forecasting, analogous to the approach described for the overall rating model. For each of the 14 tags whose statistical model beat the constant baseline, *deepseek-V3.2* was prompted with the activity title, an LLM-generated summary of the pre-activity documents, the tag label and definition, and the training-set base rate. The model was instructed to reason like a superforecaster: enumerating reasons the outcome was likely true, then likely false, then anchoring on the base rate before producing a final probability. Forecasts were produced for 150 recent training activities and all validation activities. The LLM probabilities were then blended with the statistical model predictions (80% statistical, 20% LLM). A sweep of blend weights from 0% to 70% LLM was conducted, measuring mean within-group ranking skill, accuracy, and Brier skill score across the 14 non-baseline tags on the validation set. Mean

accuracy declined monotonically with LLM weight; within-group ranking skill and Brier skill showed only marginal gains at 5–10% LLM before deteriorating; at 70% LLM the mean Brier skill score turned negative. The LLM correction was therefore discarded and the statistical model predictions are used as the final tag forecasts.

2.6.3 LLM Forecasting Methods

Several LLM forecasting methods were investigated to improve overall rating forecasts and narrative forecasting of outcomes.

In order to guard against leakage of information from the training, *deepseek-V3.2* was selected as the forecasting model for overall ratings, due to its performance on benchmarks and low cost. *gemini-2.5-pro* was also selected due to its performance. Lastly *gemini-2.5-flash* was selected only to assess the possibility of fine-tuning, as other more powerful models were unavailable for fine-tuning.

The LLM forecasting method was decided upon by iteratively inspecting both the quality of the response, and the overall accuracy of the forecasts made by the LLM. To generate the LLM forecasting methods, *gemini-2.5-flash* was prompted with a series of "mock forecasts", generated by *gemini-2.5-pro*. The "mock forecast" used relevant pages retrieved by ranking the categorized topics by forecast informativeness and retrieving 10 pages of the most relevant activity data and 10 pages of the most relevant evaluation data, prioritizing pages marked as "deviations from plans", "delays or early completion", or "over or under spending" with a minimum forecasting relevance score of 3/10, and otherwise returning the pages with the highest forecasting relevance score.

To generate each mock forecast, a retrieval-augmented input was constructed consisting of up to 10 baseline pages and up to 10 outcome/evaluation pages per activity. Baseline pages were selected from high-scoring passages in predefined "forecast-informative" categories (e.g., objectives, implementation plans, risks, financing details, contextual challenges, and stakeholder/implementer information), using a high relevance threshold (minimum categorization score of 9) and including nearby pages when insufficient high-scoring pages were available. Pages were selected from post-activity documents which emphasized deviations from plans (including deviations, delays/early completion, and over/under-spending), using a lower relevance threshold (minimum score of 3) and likewise included surrounding pages to reach the target count when needed. These retrieved excerpts were then merged with activity metadata (title, scope, planned start/end dates, planned financing totals when present) and brief model-generated baseline summaries (activity description and risk summary) before prompting Gemini to write a forecast from the ex-ante perspective. Importantly, the prompt required the model to end by outputting the *known* final evaluation rating for that activity (derived from the merged ratings file and converted into scale-specific text via `get_ratings_text`), while also instructing it to

ground the narrative in the retrieved evaluation pages and to return "NO RESPONSE" if the evaluation excerpts did not contain sufficient justification for the assigned rating.

The most semantically relevant activities which ended approximately at or before the start of the activity being forecasted were then retrieved (see Section 2.6.1. In addition, the activity "risks" were inserted before each mock forecast, to provide context for the example. Each mock forecast was structured in a way similar to the highest performing scratchpad method from (Halawi et al., 2024). A series of features including the activity title, start date, and activity location were injected into the prompt to provide context for the activity, as well as a *gemini-2.5-flash*-generated summary.

Finally, the distribution of rating outcomes was inserted into the prompt, in order to prevent collapse towards only a few ratings.

2.6.3.1 Prompt Template and Multi-stage Prompting

The full prompt is composed of the following blocks, assembled in the order listed:

1. **Activity metadata** — title, planned start/end dates, scope, planned disbursement (USD), location(s), GDP per capita, reporting organisation.
2. **Baseline description & targets** — a *gemini-2.5-flash*-generated activity description, targets summary, contextual challenge description, complexity details, integratedness description, financing summary, and implementer-performance context.
3. **Risk summary** — a *gemini-2.5-flash*-generated structured summary of key project risks.
4. **Retrieved baseline excerpts (RAG)** — additional activity-specific information retrieved by the model from the available documents and synthesised into a short passage (`{rag_synthesis_additional_info}`).
5. **KNN few-shot block** — summarised mock forecasts of the three most semantically similar training activities, one from each broad rating tier (see Section 2.6.1).
6. **Multi-stage reasoning (S1/S2)** — outputs of two prior LLM calls: reasons the activity may have been rated "Moderately Satisfactory" or worse (S1) and reasons it may have been rated "Moderately Satisfactory" or better (S2).
7. **Rating distribution prior** — a short description of the empirical distribution of rating outcomes in the training data, appended to reduce mode-collapse.

The full prompt template for the LLM Forecast is shown in Figure 2.6.3.1. LLMs in all prompts were instructed to "think like a superforecaster" (Tetlock and Gardner, 2015). Superforecasters typically outperform domain experts by maintaining broad judgement and cultivating calibrated probability estimates, rather than relying on a single strategy

such as econometric analysis. This replicates prior methods which found this instruction to be the highest performing in the geopolitical domain Halawi et al., 2024.

Few-Shot Block In both methods, a k-nearest-neighbors (KNN) few-shot block of semantically similar activities in the training data is used (see Section 2.6.1 for how semantic similarity was determined). A range of nearest neighbors was evaluated. The language model was prompted to extrapolate lessons about rating scales for the most similar "Highly Unsatisfactory", "Unsatisfactory" or "Moderately Unsatisfactory", the most similar "Moderately Satisfactory", and the most similar "Satisfactory" or "Highly Satisfactory" rated examples in the training data. A selection of $k = 3$ summarized mock forecasts was found to perform better than $k = 1$, 5, or 7.

Each example activity in the few-shot block included (i) key metadata (title and, where available, location and a brief summary), (ii) a short "risks" summary, (iii) the retrospective mock forecast analysis, and (iv) the final evaluation outcome label.

Additional Prompts Two additional prompts were given, and inserted into the final forecast: (1) reasons the activity may have been evaluated as "Moderately Satisfactory" or worse, (2) reasons the activity may have been evaluated as "Moderately Satisfactory" or better.

The forecasting prompt required a structured response format that explicitly considered arguments for both lower and higher ratings, and ended with a single-line prediction. Concretely, the model was instructed to: (1) provide reasons the overall success might be rated `{midpoint_low_text}` or lower, (2) provide reasons it might be rated `{midpoint_high_text}` or higher, (3) aggregate considerations and select exactly one of the `{num_options}` outcomes, and (4) output the final forecast on the last line beginning with `FORECAST:` followed by only the chosen option.

Finally, a short description of the empirical distribution of rating outcomes in the training data was appended. This was found to reduce mode-collapse toward a narrow subset of ratings.

Box 2.1: Single-method multi-stage forecasting prompt

Stages s1 and s2 are run as separate calls, and their outputs are inserted into the final (s3) prompt via `insert_stage_s1_answer_here` and `insert_stage_s2_answer_here`. Below are the system and user prompts.

```
SYSTEM:
You are an experienced international aid decision maker with a quantitative
mindset.  Respond with a comprehensive, thorough forecast of what the overall
evaluation rating of the activity will be, from the options of {options_text}.

USER:
Forecast what the outcome will be for this activity.
```

```
### Lessons from similar activities ###
{knn_summary_text}
### End lessons ###

### Additional specific information about the activity that you summarized ###
{rag_synthesis_additional_info}
### End of additional information you summarized ###

ACTIVITY ID: {activity_id}
ACTIVITY TITLE: {activity_title}
ORIGINAL PLANNED START DATE: {planned_start}
ORIGINAL PLANNED END DATE: {planned_end}
ACTIVITY SCOPE: {activity_scope}
PLANNED TOTAL DISBURSEMENT (USD): {planned_total_disbursement_usd}
ACTIVITY LOCATION(S): {locations}
LOCATION GDP PER CAPITA, USD: {gdp_percap}
PARTICIPATING ORGANIZATIONS: {reporting_orgs}
IMPLEMENTING ORGANIZATION CATEGORY: {either "Government" or "NGO", otherwise
line not inserted}

ACTIVITY DESCRIPTION: {ChatGPT_description}
ACTIVITY TARGETS: {targets_summary}
ACTIVITY CONTEXT: {activity_context}
ACTIVITY COMPLEXITY: {complexity_details}
ACTIVITY INTEGRATEDNESS: {how_integrated_description}
FINANCING DETAILS: {finance_summary}
IMPLEMENTER PERFORMANCE CONTEXT: {implementer_performance_text}
ACTIVITY RISKS:
{risks_summary}
ACTIVITY POSSIBILITIES: {possibilities_summary}

{training-set rating distribution text}
Here are a few reasons that you said the answer might be "Moderately
Satisfactory" or worse:
{insert_stage_s1_answer_here}
Here are a few reasons that you said the answer might be "Satisfactory" or
better:
{insert_stage_s2_answer_here}

YOUR TASK:
Aggregate your considerations above.  Think like a superforecaster (e.g. Nate
Silver).  On the very last line of your response, write 'FORECAST: ' followed by
exactly one option from this rating scale with no extra words:
{options_text}

Respond only in English.
```

2.6.3.2 Inference-time Enhancements (Ensembling)

Ensembling averages many individual forecasts of the same model, prompted in slightly different ways. Ensembling was found to be a weak method when running on 100 sample activities within the training set, because the same model+prompt pair had highly correlated forecasts. Creating a new prompt variant for each ensemble run was also not attempted due to the complexity of such an approach, its high API cost, and weak expected performance improvements.

2.6.3.3 Model Adaptation (Fine-tuning / DPO)

In past work in a similar domain, fine-tuning significantly improved forecasting performance (Wen et al., 2025). An attempt was made to fine-tune *gemini-2.5-flash* using Vertex AI. To do so, Direct Preference Optimization (DPO) was used, which requires one example of a good prompt, and one example of a bad prompt. Out of a random sample of 100 activity IDs in the training data, the final forecast stage was forecasted 5 separate times using *deepseek-V3.2*. 50 pairs were used where the model forecasted one rating increment closer than another rating, as the pairs of forecasts.

It was often the case that there were multiple options for the best or worse forecast due to the limited 6-point scale. In order to choose among forecasts that were equally good or equally bad, the embeddings of the forecasts were also taken and cosine similarity was computed to the embeddings of the mock forecast, and to embeddings of the post-activity document, and these similarity scores were averaged. The most similar among equally good ratings were chosen as the good example for the DPO training pair, and the least similar among equally bad ratings as the bad example.

Once the 50 pairs were identified, fine-tuning was conducted using default settings over 20 epochs from Vertex AI (Learning rate multiplier of 1, Adapter size of 4, Beta of 0.1).

2.6.4 LLM Adjustment Ridge Regression

The goal of the LLM Adjustment was to incorporate usable information from the direct LLM forecasts, and because simply averaging the LLM forecasts with the statistical model in the validation set decreased performance. To do so, a small ridge regression model was layered on top of the forecasts, calibrated using LLM predictions on the training set.

Let $\hat{y}_i^{\mathrm{RF}}$ be the RF+ET prediction for activity i, and let $\hat{y}_i^{\mathrm{LF}}$ denote the LLM Forecast. Note that the RF prediction was also averaged with the ET model, but it is still referred to as "RF" in the equations. The RF+ET residual on the i'th activity is defined as:

$$r_i := y_i - \hat{y}_i^{\mathrm{RF}}.$$

A ridge regression model was then fitted to predict residuals from a small feature vector consisting of the RF+ET prediction and (optionally) the LLM Forecast:

$$\hat{r}_i := \beta_0 + \beta_1 \hat{y}_i^{\mathrm{RF}} + \beta_2 \hat{y}_i^{\mathrm{LF}},$$

with an ℓ_2 penalty on (β_1, β_2) controlled by $\alpha = 5$, the ridge strength. The corrected prediction is:

$$\hat{y}_i^{\mathrm{corr}} := \mathrm{clip}_{[0,5]}\left(\hat{y}_i^{\mathrm{RF}} + \lambda \hat{r}_i\right),$$

where λ is a scaling factor (set to 1.0 in the experiments) and clipping enforces the valid

rating range between 1 and 6.

The ridge model takes both the RF+ET prediction and the LLM Forecast as inputs for activities where $\hat{y}_i^{\mathrm{LF}}$ is available (the training activities for which LLM forecasts were generated, spanning the full validation set. This allows the correction to learn a mapping from $(\hat{y}_i^{\mathrm{RF}}, \hat{y}_i^{\mathrm{LF}})$ to the residual r_i. The ridge model under this setting effectively learns how much LLM forecasts should shift the prediction from the mean, how much the RF+ET model should shift the prediction from the mean, as well as a fixed learned offset. The only learned parameters are β_0, β_1 and β_2.

2.7 Scoring Metrics

Accuracy The percent of the time the correct rating is forecasted. Non-integers are rounded to integers for overall ratings. While accuracy has the advantage of intuitive simplicity, is not strictly proper (Gneiting and Raftery, 2007), meaning that maximizing the model accuracy may not maximize a model's reported true belief about the probability of an outcome. This can be problematic when being used to select between models.

RMSE (Root Mean Square Error) Take the square of the difference between every prediction and the true value, take the mean of all such squared values, then take the square root. Measure of "average" distance. Lower is better. As ratings range from 1 to 6, the difference is on a scale from 0 to 5, where the worst possible value is 5, best possible value is zero. This method heavily penalizes predictions that are significantly incorrect. RMSE is only used in this work for overall ratings.

MAE (Mean Absolute Error) Measure of "average" distance, by taking the mean of the absolute value of the residuals. Lower is better. As ratings range from 1 to 6, a scale from 0 to 5 is best, therefore worst possible value is 5, best possible value is zero. In the case of overall success ratings, this metric tends to reward integer rounded rating scores as they are often exactly correct, rather than the output of the statistical models which are not exact integers. This method does not heavily penalize predictions that are significantly incorrect. MAE is only used in this work for overall success ratings.

Coefficient of Determination (R^2) R^2: Coefficient of determination. Theoretically equals zero, if we always choose the mean (however using the training set mean results in a lower score on the test set in the baseline measure below). If more than 1 regressors are included, R^2 is the square of the coefficient of multiple correlation and can be negative. Measures proportion of the variation in the dependent variable that is predictable from the independent variable. Higher is better. This method generally does not penalize outliers significantly. R^2 is only used in this work for overall ratings and cost-effectiveness.

Adjusted R^2 Adjusted R^2 is a version of R^2 that accounts for the number of regressors in the model, and is well-defined for OLS (Ordinary Least Squares) linear regression. Unlike

plain R^2, adjusted R^2 penalizes adding predictors that do not meaningfully improve fit. It can decrease when irrelevant regressors are included. A higher adjusted R^2 is better. While it penalizes extra parameters that may lead to overfitting, adjusted R^2 within a training set does not reflect model skill as accurately as out-of-time R^2. Adjusted R^2 is reported in this work on the training+validation set only for comparability with prior literature, while test set R^2 and within-group pairwise ranking are the primary results. Adjusted R^2 is only used in this work for overall ratings.

Within-Group Pairwise Ranking Within-group pairwise ranking is evaluated as the mean across groups of the proportion of within-group pairs of predictions that were correctly ordered. Only pairs of activities sharing the same reporting organisation and start year are compared; between-group pairs are excluded entirely. For each such pair (i, j) where the true outcomes differ ($y_i \neq y_j$), the prediction is counted as correct if $\hat{p}_i > \hat{p}_j$ whenever $y_i > y_j$, and incorrect otherwise; pairs where $\hat{p}_i = \hat{p}_j$ are excluded from both numerator and denominator. Each group's score is computed independently and the final metric is the weighted mean across groups proportional to the number of activities in each group, so a large group does not dominate a small one. Restricting comparison to activities from the same organisation and group year removes systematic between-group variation in base rates that a model cannot be expected to predict, making the metric a cleaner measure of within-context discriminative skill. This method is insensitive to global shifts in ratings or probabilities, which may occur due to events like the COVID-19 pandemic or differences in organisational reporting culture, and insensitive to calibration of the spread of predicted probabilities. A score of 0.5 corresponds to random chance; 1.0 is perfect ordering.

Within-Group Spearman Ranking Within-group Spearman rank correlation measures the monotonic association between predicted probabilities and true outcomes sharing the same reporting organisation and start year. For each group with at least two observations, Spearman's ρ is computed by assigning midranks to both the true outcomes and the predicted probabilities (so that tied values receive the average of the ranks they would have occupied), then computing Pearson's r on the two rank vectors. The use of midranks rather than the shortcut formula $1 - 6\sum d^2/n(n^2 - 1)$ is necessary because the shortcut assumes all ranks are distinct integers; with ties the identity it rests on breaks down and values outside $[-1, +1]$ become possible. The reported score is the weighted average of per-group ρ values, with weights proportional to group size. Like within-group pairwise ranking, this metric controls for between-group differences in base rates, but additionally captures the degree of monotonic ordering rather than merely whether the direction of each pair is correct.

Brier Score The Brier score, used here in its binary form, is the mean squared error

between the predicted probability $\hat{p}_i$ and the binary outcome $y_i \in \{0, 1\}$:

$$\mathrm{BS} = \frac{1}{N} \sum_{i=1}^{N} (\hat{p}_i - y_i)^2.$$

Lower values are better, with 0 indicating perfect probabilistic forecasts and 1 being worst possible. The Brier score is a strictly proper scoring rule (Gneiting and Raftery, 2007), meaning that a forecaster maximises their expected score only by reporting their true beliefs. Only the binary Brier score is used in this work for outcome tags.

Brier Skill Score The Brier skill score (BSS) measures improvement over a reference forecast:

$$\mathrm{BSS} = 1 - \frac{\mathrm{BS}}{\mathrm{BS}_{\mathrm{ref}}},$$

where $\mathrm{BS}_{\mathrm{ref}}$ is the Brier score of the constant (no-skill) baseline that always predicts the training-set base rate. A BSS of 0 indicates no improvement over this baseline; positive values indicate skill. Brier skill score is only used for outcome tags.

2.7.1 Choosing Methods and Metrics for the Held-out Test Set

Rating and Cost Effectiveness Evaluation After consideration of several metrics, the within-group pairwise ranking and R^2 were determined to be the most appropriate for assessing model skill in the domain of ratings and activity quantitative outcomes, while other metrics still provide useful insight. R^2 is sensitive to bias and outliers, which are important for assessing absolute predictive accuracy. However, a common use-case in aid funding decision making is comparing a pair of activities, or even a suite of many activities. In such a use-case, the within-group pairwise ranking is more representative of what is needed. Furthermore, a weakness in R^2 is that global shifts in the ratings and outcomes due to external factors can be unpredictable, and a model may by-chance capture these shifts without any real improvement in forecasting skill. Both are clearly interpretable: always forecasting the mean value of the held-out set is equivalent to an R^2 of 0, also meaning that 0% of variance explained for R^2, while 50% is equivalent to the within-group pairwise ranking forecast of random chance.

Methods chosen for the Held-out Test Set Even after rigorously evaluating the methods which worked well on the validation set, I wanted to evaluate whether some methods offered performance improvements on the test set. The selected methods to evaluate were as follows:

- **Statistical Models for Overall Ratings:** As baseline methods I chose to assess the performance of always predicting the most common rating within the reporting org (*Mode of reporting-org score baseline*), a simple ridge regression model with

only the overall risk LLM grade and the reporting org (*Ridge Baseline (risks + org only)*) as a means of approximating what human evaluators may have assigned the likely overall rating, and a simple linear model with all features (*Plain OLS GLM*) to assess the importance of nonlinearity for the statistical model performance on the test set. My primary method was (*RF+ET all features*) which was the average of the RF and ET model after adjusting year correlation, and I assessed the following alterations to see if they improved performance: the RF+ET ensemble averaged with the LLM forecast (*RF+ET + LLM Forecast (simple avg)*), the more sophisticated calibrated ridge regression to combine the statistical model with the LLM forecast *RF+ET + LLM Forecast (ridge)*, removing the linear year correlation adjustment (*RF+ET all features - year corr*), removing LLM input features from the model (*RF+ET, no LLM features*), resetting the RF+ET tuned parameters to their default values (*RF+ET (default params)*), and replacing the RF+ET with the XGBoost nonlinear model (*XGBoost all features*).

- **Outcome Tags** I assessed only two methods on the test set for outcome tags: including the per-tag features and parameters (all subsets of the 45 "non-noisy" features, where each per-tag parameter set was a less complex model), and compared this with using only the same 45 "non-noisy" features which did well on the validation set.

2.7.2 Grading Free Form Forecasts

In addition to extracting the LLM forecasted rating, *gemini-2.5-flash-lite* is also used to grade narrative textual format forecasts. In order to do so, *gemini-2.5-flash* was first used to summarize pages categorized as highly relevant to post-activity evaluations, obtaining 10 pages with a score of at least 3/10 marked as "deviation_from_plans", "delays_or_early_completion", or "over_or_under_spending", or failing that, that were graded as highly relevant to activity evaluation. If no such pages were categorized, the first 10 pages of the highest ranked activity post-activity evaluation were used. Next, for all ensembles of LLM forecasts, a grade was given for how similar the forecast was to the activity outcome using *gemini-2.5-flash-lite*. Grading was performed based on two key criteria: accurately identification of likely drivers of activity success or failure, and identification of likely outcomes being forecasted, which together inform the single combined letter grade.

In accordance with the typical US grading scheme definitions, an "F" grade is defined as 55 or lower and an A+ grade as approximately 97, with other grades defined in even intervals. *gemini-2.5-flash-lite* was provided the following rubric:

Grading scale

- A+/A/A-: Excellent forecast, highly accurate, attention on key drivers, multiple major events forecasted accurately.
- B+/B/B-: Good forecast, mostly accurate. A mix of correct and incorrect, but at least one major outcome was forecasted. At least one key driver identified.
- C+/C/C-: Adequate forecast, partially accurate or reasonable. Focus was adequate. Major outcomes incorrect, but some smaller aspects were correct.
- D+/D/D-: Poor forecast, although perhaps one or two small correct things. Mostly inaccurate. Wrong focus on drivers.
- F: Failed forecast, completely wrong or unsupported.

2.8 Summary

This thesis implements four forecasting strategies, all trained and evaluated on a dataset of 1,181 completed international aid activities from the IATI database:

- A statistical model to forecast overall activity success ratings using LLM-generated features
- A statistical model to forecast activity cost-effectiveness using LLM-generated features
- A statistical model to forecast 14 binary outcomes using LLM-generated features
- An LLM prompting scheme used to forecast overall activity success ratings, assessed on accuracy, within-group ranking skill, and qualitative similarity to the evaluation summaries, optionally incorporating statistical overall rating forecasts as a target.

The system is built to forecast what the evaluation results will be for thousands of IATI records containing both a pre-intervention description of the activity, and an ex-post evaluation of the results in order to ensure temporal leakage does not occur. To do so, thousands of PDFs were downloaded, ranked from most to least relevant for forecasting future outcomes or evaluating the end result of the activities, had their paged graded for relevance to the task, and had quantitative and qualitative descriptions and results extracted from the pdfs. Three time-ordered splits were used: train, validation, and test sets. Statistical models and fine-tuning experiments were done on the training set, with several methods assessed using the validation set. This informed model selection and parameter tuning, so that the generalization ability of these methods were finally assessed against the held-out test set. See Figure 5 for a representation of the entire process.

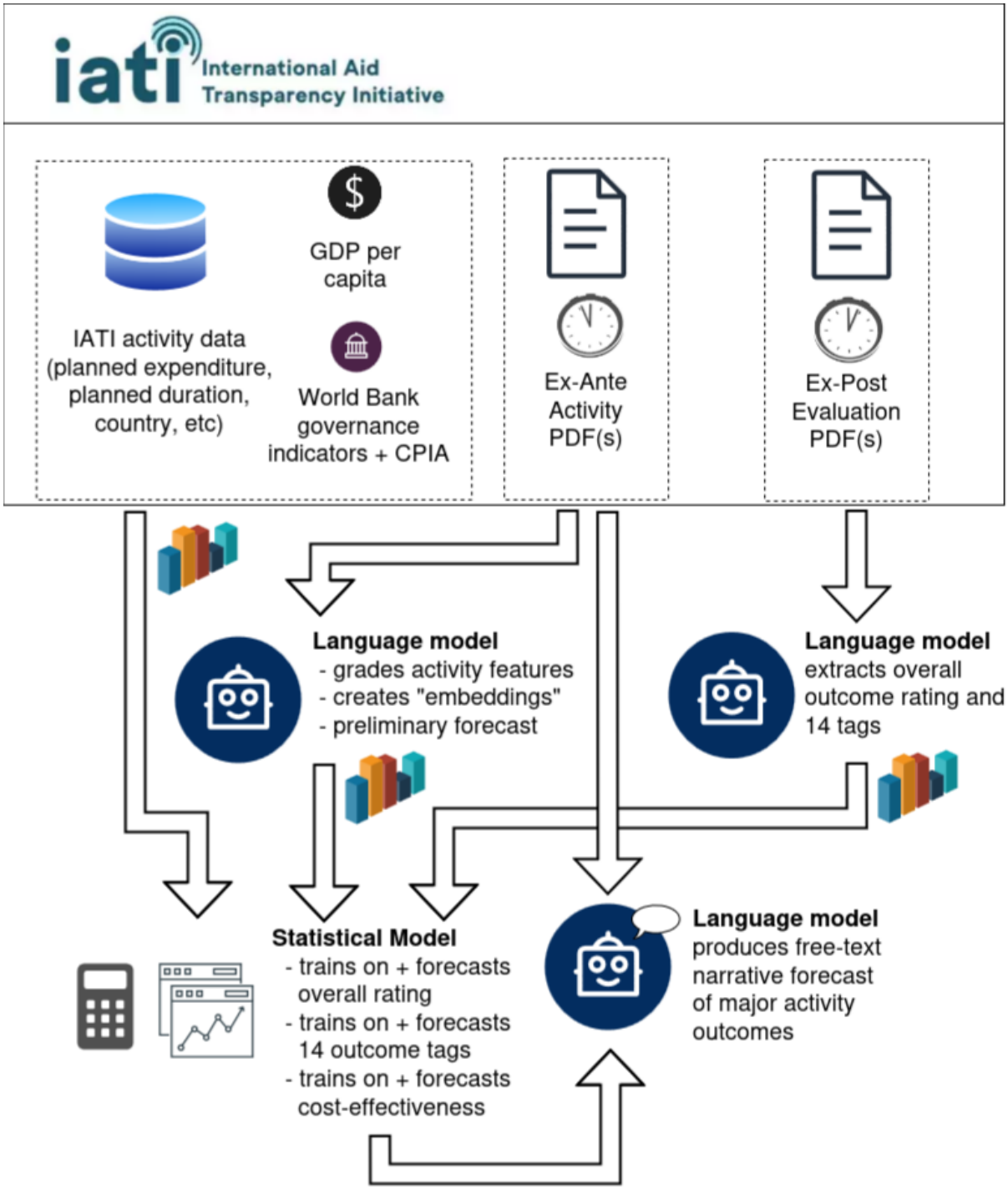

Figure 5: Overview of the forecasting pipeline. Features are produced using IATI activity data and external by-country datasources (GDP per capita, CPIA scores, and World Bank governance indicators). More features come from ex-ante activity PDFs processed by *gemini-2.5-flash* into structured data. Lastly, *ChatGPT-3.5* generated more features by grading them directly, and *gemini-embedding-001* was used to create more features using semantic embeddings. Next, *deepseek-V3.2* was used to produce a preliminary forecast for overall ratings and outcome tags. A statistical model also trains on and forecasts overall ratings, 14 outcome tags, and cost-effectiveness, optionally using the *deepseek-V3.2* forecasts as input features. *gemini-2.5-flash* processes ex-post evaluation PDFs to extract the overall outcome rating and 14 tags, which serve as training targets for the statistical models. Lastly, language model forecasting is qualitatively evaluated:

3 Results

3.1 Summary

The sections below present results for four outcome measures: overall evaluation ratings, narrative forecast similarity grades, aggregate cost-effectiveness Z-scores, and structured outcome tags. Figure 7 provides the central empirical comparison across LLM forecasting methods and constitutes the primary evidence for the findings discussed in Section 3.4.

Four methods for measuring activity outcomes were compared.

Activity ratings results Overall rating results are shown in Table 1. The overall activity success rating prediction on the held-out set confirmed the inability to explain variation in evaluation ratings in absolute terms which was evident from cross-validation tests on the validation set. As expected, within-group pairwise ranking ability on the held-out set was above chance and statistically significant. The use of LLM features improved all metrics, as did the hyperparameter tuning of the RF+ET model parameters. It was also revealed that nonlinear methods are essential for within-group pairwise ranking skill, with the OLS model both less accurate than the baseline and failing to rank within-group pairs.

However, not all experimental methods succeeded in improving the model. Inspection of the LLM forecasts on the test set revealed significant future leakage based on the reasoning the LLM gave for the ratings, so LLM-forecasting combinations with the test set should be understood to have an unfair forecasting advantage. Furthermore, the ridge regression method to calibrate LLM forecasts backfired on the test set, performing worse than a simple average of the LLM forecast with the RF+ET prediction. The temporal linear regression produced mixed results, decreasing overall accuracy, but improving within-group pairwise ranking and within-group spearman ranking ability.

LLM prediction of overall ratings LLM predictions on the held-out set were better than the statistical model, but were discovered to be strongly affected by future leakage. The validation set results did not demonstrate clear future leakage, and were inferior to statistical model predictions. A fusion of statistical models and LLM forecasting methods was found to generate the highest quality narrative predictions. The highest rating accuracy for LLM forecasts simultaneously produced the least accurate narratives as shown in Figure 7.

Cost-effectiveness Activity ratings show only weak correlation with activity aggregate cost-effectiveness (Pearson $r = 0.07$, computed across the full dataset of 1,181 activities). Cost-effectiveness prediction was not shown to be statistically better than chance with the 95% permutation sampling for any scoring metric.

Outcome tags Outcome tag results are shown in Table 3. Outcome tags had a similar

within-group ranking skill as the overall rating model, with some notably stronger performing outcome tags. The regularization methods for outcome tags overall decreased forecasting skill compared to a minimal regularization. Results are reported for a minimal, uniform regularization rather than the hand-selected feature sets from the validation set. This is because of the two methods attempted on the held-out set, the less tuned feature selection performed better, although the performance difference was modest.

Across 14 binary outcome tags, including activity rescoping, finance and budget, and target achievement, the RF+ET average with no per-tag feature selection achieved an average within-group pairwise ranking of 59% (range: 47% to 71%) for the 45% of pairs with differing outcomes, outperforming the alternative model which used feature selection. By comparison, the per-tag feature selection model achieved an average of 57% (range: 48% to 71%).

The results from each method are described in detail in the remaining sections.

3.2 Temporal Information Leakage

The test set had a high rate of apparent leakage - direct inspection revealed several clear cases. It is hypothesized this is due to the high influence of COVID and conflict, and general upheaval in 2019 which especially affected activities starting between 2016 and 2020. Therefore, while the test set results from language models perform well, this should be considered untrustworthy and understood to have a clear unfair advantage compared to a real forecast or the validation set. The validation set did not show significant leakage concern. Manual inspection did not reveal any leakage. A script was written to search for common keywords such as COVID or "Ukraine war" discovered that 4/298 val activities have COVID leakage (all started 2015–2016, before Jan 2020), and no Ukraine war matches.

3.3 Forecasting Ratings with Statistical Models

Overall, the forecasting system developed is capable of forecasting evaluation ratings significantly above chance on out-of-time activities at 95% confidence, with the RF+ET achieving a pairwise ranking skill of 60 % [95% CI: 53 %, 66 %]. This can be contrasted with the R^2 on the test set of -0.002, although it is still well above the baseline R^2 of -0.05from the ridge regression.

Despite its low accuracy on the test set, the model performs similarly to published literature within the training and validation set. The full RF+ET model has an R^2 on the combined training and validation set of 0.32. Existing literature as of 2021 reports at maximum 30% of unexplained variance on the World Bank using linear OLS models (L.

Ashton et al., 2023). An OLS linear model trained on all features in the final 4 reporting organization dataset after applying the mode-demeaned fixed effect for the training and validation set shows an R^2 of 0.18 and an adjusted R^2 of 0.13, which is in the expected range when the fact that no information available at evaluation such as the final duration and expenditure of the program was used. No comparable out-of-time, time-ordered split analysis was identified in the literature. As expected for forecasting under data distribution shift, results on the out-of-time validation set were significantly weaker, This is likely due to significant structural shift over time in evaluation rating practices and increased influence of external factors such as the COVID-19 pandemic in the test set period.

Several factors were found to improve model performance. As the ET model performed similarly to the RF model with a slightly different algorithm, it was found that averaging the random forest model with the extra trees model also improved forecasting performance. Further successful techniques are listed below.

3.3.1 Overfitting Corrections

"Adjusted R^2" has been used in similar work to evaluate model performance in the development aid literature and penalizes overfitting by reducing the reported R^2 as a function of the number of input parameters. Mirroring similar reported methods in the literature, adjusted R^2 was calculated on the training points, in order to compare the model with results from prior literature. Because nonlinear models were found to be significantly superior, other methods were required to prevent overfitting and aid out-of-time generalization. Adjustments to RF+ET parameters were found to be strongly influential for model performance.

3.3.2 Language Model Features Used for the Statistical Model

Adding the language model derived features including LLM grades (finance, integratedness, implementer_performance, targets, context, risks, and complexity), embedding features (UMAP X, Y, and Z), and sector cluster allocations to the RF raises validation R^2 from -0.026 to -0.002 ($\Delta R^2 = 0.024$; see Table 1, rows "Random Forest, no LLM features" vs. "Random Forest only").

Table 1: Held-out test set performance in forecasting ratings across forecasting methods. Rows are sorted by ascending R^2 (higher is better). RMSE and MAE are lower-is-better, others are better if higher. Bold indicates the best value in each metric column. It was noted that MAE was better when the metric was rounded, so rounding the best model was attempted, which was found to beat the baseline metric on MAE. This is because the activity success ratings are integers, but the statistical models produce continuous prediction. The "LLM correction" variants combine models using the 300 latest-starting activities in the validation set for calibration/combination using the ridge regression model. XGBoost was found to underperform the RF when used as the LLM Forecast base. The Pairwise metric is artificially suppressed for the "Mode of rep. org. score baseline". The "no LLM features" RF excludes the following features: finance, integratedness, implementer_performance, targets, context, risks, complexity, umap3_x, umap3_y, umap3_z, and the 15 sector allocation features. Note: rows shown in orange include LLM Forecast predictions, which were found to benefit from future leakage and should be treated cautiously.

Method	R^2 ↑	RMSE ↓	MAE ↓	Acc. ↑	WG Spear. ↑	WG Pair. Rank. ↑
RF+ET + LLM Forecast (simple avg)	**0.004**	**0.924**	0.683	0.470	0.206	**0.599**
RF+ET all features (no year corr)	**0.004**	**0.924**	0.683	0.470	0.206	**0.599**
RF+ET + LLM Forecast (ridge)	-0.002	0.927	0.704	**0.485**	0.206	**0.599**
RF+ET all features + year corr	-0.002	0.927	0.704	**0.485**	0.206	**0.599**
XGBoost all features	-0.008	0.929	0.688	0.460	0.174	0.580
RF+ET, no LLM features	-0.026	0.938	0.680	0.475	0.070	0.541
RF+ET (default params)	-0.036	0.942	0.706	0.465	**0.206**	0.597
Ridge Baseline (risks + org only)	-0.048	0.948	0.696	0.465	0.008	0.508
Mode of reporting-org score baseline	-0.079	0.962	**0.658**	0.460		
Plain OLS GLM	-0.136	0.987	0.767	0.410	0.054	0.529

3.3.3 Explaining Feature Importances

SHAP (SHapley Additive exPlanations) is used to interpret model predictions by attributing changes in the output to individual features using Shapley values. For each observation, SHAP decomposes the prediction into a baseline (the mean model output) plus additive feature contributions. Each feature's SHAP value represents its average marginal contribution across all possible feature subsets.

SHAP analysis reveals that increased planned expenditure and decreased duration strongly correlate with activity success. The reporting organization dummy variable—in this case identifying World Bank activities, which have below-average ratings in this dataset—was among the most important features. The urban/rural indicator from the embeddings tended to decrease ratings, weakly indicating that less well contextualized activities performed worse. Contextual challenge weakly shifted ratings downwards, while ease of targets shifted predictions upwards. Projects spanning many countries tended to perform slightly worse.

3.4 Forecasting Ratings and Narrative Forecasts with LLMs

Figure 7 summarises performance across all LLM forecasting configurations. The best narrative similarity grade is 88.5 (a "B+", corresponding to forecasts that are on average between "mostly accurate" and "highly accurate"), achieved by the RF-forced method with full context (RAG+KNN+S1+S2). The minimal-prompt baseline achieves the highest R^2 for rating prediction but the worst narrative similarity, while RF-forced methods dominate on both narrative grade and within-group pairwise ranking probability.

One research question of this thesis is how do differing methods of LLM forecasting apply in the context of forecasting international aid success ratings. More generally capable models were found to be better forecasters. The LLM's ability to predict the rating was also found not to clearly correlate with the quality of its forecast narrative. In fact, the best rating accuracy from an LLM was associated with the least accurate stated reasons. The strategy of incorporating additional information relevant to the forecasting question as well as references to similar activities does not clearly improve forecasting skill.

Inspection did not reveal significant information leakage on the validation set, but considerations of the impacts of COVID and other events were prevalent in the test set. Indeed, a test of *gpt-3.5-turbo* and *gpt-4* with the same prompt as *deepseek-V3.2* revealed that the absolute error increased significantly on average after the model cutoff date relative to the RF+ET model. While *gemini-3-pro* and *deepseek-V3.2* with thinking enabled performed better than the RF+ET ensemble on within-group pairwise ranking, this is likely at least in part due to information leakage. Future work should test these more powerful models against larger ensembles of results after their cutoff dates, to provide a fair test against

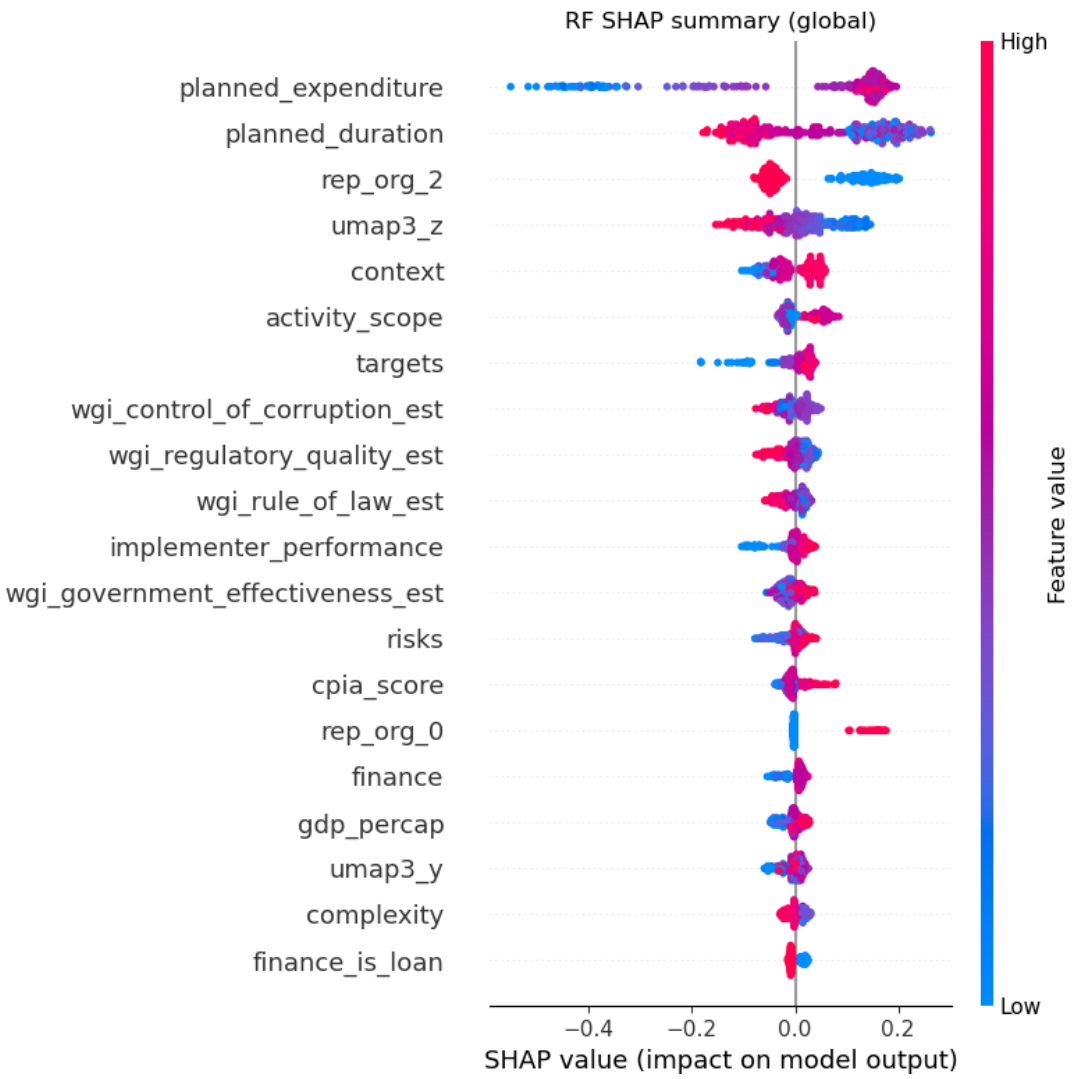

Figure 6: A SHAP analysis of ratings on the validation set from the RF. Red indicates an increase in the value of the feature, while blue indicates a below-average value. Points to the right of zero shift ratings up, points to the left of zero shift ratings down.

statistical models which were rigorously vetted to not contain information leakage.

Little evidence was found that fine-tuning can improve forecasting accuracy in this domain, although a thorough assessment of the potential for fine-tuning *gemini-2.5-flash* was cost prohibitive. Language models on their own were also found to consistently underperform statistical models in forecasting skill, with the exception of within-group pairwise ranking ability from *deepseek-V3.2* with reasoning enabled. However, GPT-3.5, GPT-4, and *deepseek-V3.2* with reasoning turned off all performed worse than the RF model on the test set items, and even *deepseek-V3.2* performed worse on absolute accuracy than the RF+ET model in the test set, indicating that the superior performance was likely due to chance. Currently, language models have a high forecasting variability between models and prompts. Using statistical model prediction as the target for the LLM to forecast (e.g. "Conclude with a forecast of *moderately satisfactory*" as derived from a statistical model

prediction) was found to significantly improve the quality of LLM narrative forecasts.

In addition to extracting the LLM forecasted rating, LLM grades were also assessed. The assessed similarity between forecasts and summaries of post-activity evaluations was found to be usually correlated with improved ability to predict ratings, with the exception that very low context prompts are able to forecast accurately but for the wrong reasons (the "Baseline" method) (see Figure 7). Furthermore, it was found that when the predicted rating using the best statistical model is injected into the prompt, LLM forecasts of activities become much more similar to the summaries of their post-activity evaluations. This provides evidence that while language models can accurately forecast the reasons activities would succeed or fail if given the evaluation ratings, they generally lack the ability to correctly weigh the various factors to produce a calibrated forecast. We should be wary about using rating accuracy prediction as a sole benchmark for LLMs, given that the LLM prompt with the worst similarity to the actual rating also had the highest forecasting success. This is likely due to increased model ability when the context window is not as full - the model is better at forecasting when not overwhelmed with information in its context window, but lacking that key information worsens its ability to pick out the best reasons for the forecasts. Another option could be that the model focuses on what it does know, such as knowledge of supply chain disruptions or conflict occurring during that timeframe in the host country.

While ratings are not clearly correlated with better similarity grades, RAG, KNN retrieval, and extra reasoning of the models (stages 1 and 2) consistently improve narrative textual similarity to the post-activity summary, albeit by a small margin. Simply presenting the grading model with the *gemini-2.5-flash* summary of risks of the project from the project documents lead to a forecasting score of 80.2 (between a "C+" and a "B-", corresponding to an "Adequate" forecast, with only "Adequate", rather than a good, focus on the drivers). However, the risks did not typically include any affirmative forecasts of post-activity outcomes, which may in part explain the low ratings. The best forecasting method in terms of similarity to the post-activity document summary was the forced ratings with a large amount of context (RAG+KNN) and included stage 1 and 2, achieving a grade of 88.5, which corresponds to a "B+" grade given by *gemini-2.5-flash-lite* (which sits midway between "Good" and "Excellent", meaning the forecasts were on average between "mostly accurate" and "highly accurate", with at least one major outcome forecasted, and at least one key driver of the outcome identified on average). Without any additional methods, *deepseek-V3.2* can typically achieve a "B" grade on average, while use of the statistical model or of *gemini-3-pro* can lift scores to the "B+" range. The improvement in ability to forecast overall ranking improves significantly as well.

Fine Tuning The training loss for *gemini-2.5-flash* was found to have steadily reduced over the course of the fine-tuning, reducing by 95%. This was tested on an early subset of the validation data which excluded BMZ activities. The performance results were mixed.

While the R^2 performance improved modestly, it remained worse than simply guessing the mean rating in the validation set, and remained far below the performance of the more powerful *gemini-3-pro*. Furthermore, the DPO objective is specifically intended to improve within-group pairwise ranking performance. However, the within-group pairwise ranking was slightly worsened by fine-tuning. The cost for the fine-tuning of 20 epochs for 50 pairs was close to 140 euros. Therefore, scaling up larger training points for fine-tuning does not seem to be a promising strategy, either to reach the performance of *gemini-3-pro*, or to reach the performance of the more powerful combined statistical models.

RAG and KNN Small differences in forecasting skill were found for the addition of RAG and KNN on forecasting ratings. However, the narrative similarity scores consistently improve on the validation set when RAG and KNN context is included. This indicates that while the LLM is able to identify more important information when reasoning about the final forecast, it is insufficiently calibrated to use the additional information it has gathered to improve the forecast itself.

Forecasting LLM Model Selection No consistent difference was found between *gemini-2.5-flash* and *deepseek-V3.2* in forecasting skill. However, *gemini-3-pro* appears to be significantly better at forecasting than both, in both sets of activities achieving higher similarity grades, higher R^2, and higher within-group pairwise ranking than other models, in both the validation and the test set. Two exceptions apply. Models instructed to match the statistical model's forecast outperform *gemini-3-pro* on similarity grades. Additionally, *deepseek-V3.2* using only the minimal prompt achieves a modestly better R^2 and within-group pairwise ranking for ratings on the validation set, while scoring a much worse forecast similarity score. When compared directly on the test set using the same prompt *gemini-3-pro* performs better than deepseek.

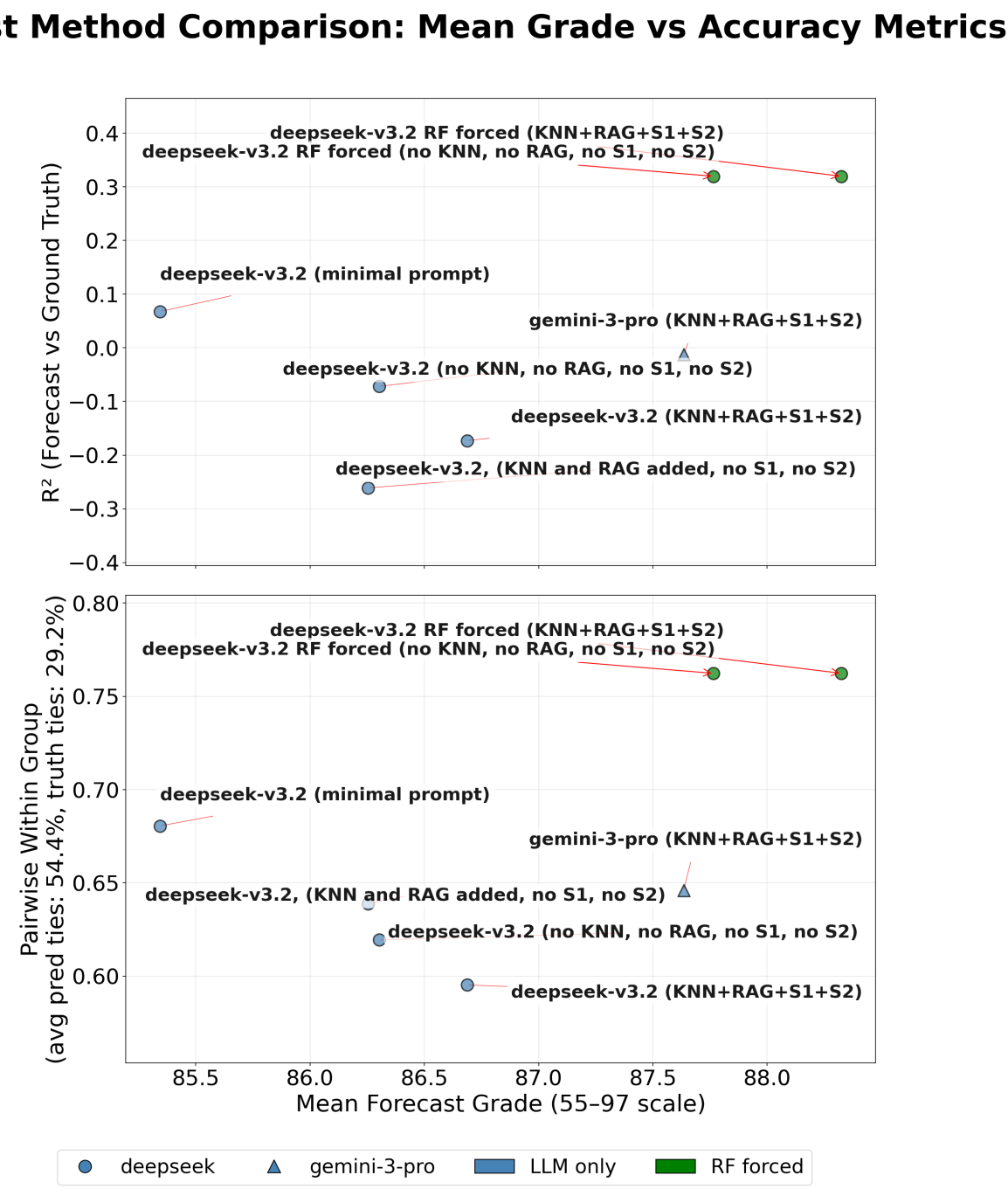

Figure 7: A comparison of LLM forecasting methods for the validation set. The x-axis shows the mean narrative forecast grade on a scale from 55 to 97 (higher is better; graded by *gemini-2.5-flash-lite* against the post-activity document summary). The y-axis in the top row shows R^2 between predicted and true ratings (higher is better; negative values indicate worse-than-mean predictions). The y-axis in the bottom row shows within-group pairwise ranking (higher is better; 0.5 = chance). No confidence intervals are shown for individual LLM configurations. The main finding is that RF-forced methods (green) dominate on narrative grade and pairwise ranking, while the minimal-prompt baseline achieves the highest R^2 illustrating a tension between rating accuracy and narrative quality. 288 of the 300 validation set activities with overall ratings had sufficiently detailed outcome evaluations to produce a summary. "KNN" refers to the retrieval of 3 similar examples into the prompt, "RAG" refers to the model additionally retrieving information it decided it was missing from the available documents and adding to the prompt, "S1" refers to additional reasoning injected into the context considering why the activity outcome may be overall unsatisfactory, "S2" refers to additional reasoning in the context considering why the activity outcome may be overall satisfactory.

Table 2: Random-forest drop-one feature importance, sorted by absolute change in the prediction from the feature being increased by 1 standard deviation (Δ `pred_1sd`). Each row reports the decrease in validation R^2 and pairwise ranking when the feature is removed (ΔR^2), along with the share of training rows that were median-imputed for that feature (`Miss` %). The largest impacts come from planned budget and planned duration.

Feature	Miss %	$\Delta R^2 \uparrow$	ΔPairwise $\uparrow$	Δpred_1sd
Feature	Miss %	$\Delta R^2 \uparrow$	ΔPairwise $\uparrow$	Δpred_1sd
Target achievability (LLM: 100=easy, 0=impossible)	2.0	0.0077	0.0245	0.0920
Document embedding Z axis	8.5	0.0076	0.0115	-0.0672
External context (LLM: 100=enabling, 0=hostile)	1.5	0.0023	0.0038	0.0565
Sector: capacity building	47.0	0.0026	0.0017	-0.0456
Region: East Asia & Pacific	0.0	0.0132	0.0309	0.0381
Implementer quality (LLM: 100=excellent, 0=poor)	0.5	0.0092	0.0162	0.0308
Annual expenditure (log)	8.5	-0.0009	-0.0076	0.0254
Document embedding Y axis	8.5	0.0025	0.0029	0.0218
Reporting org: World Bank	0.0	0.0285	-0.0059	0.0217
Planned duration	0.5	-0.0001	-0.0018	-0.0216
Expenditure x complexity	7.0	0.0032	0.0070	0.0210
Document embedding X axis	8.5	0.0034	0.0083	-0.0180
Geographic scope	0.0	0.0007	-0.0127	-0.0175
Finance adequacy (LLM: 100=well-funded, 0=underfunded)	4.5	0.0074	0.0131	0.0171
Region: Middle East & North Africa	0.0	0.0013	0.0052	-0.0160
Country governance quality (CPIA)	31.5	-0.0001	-0.0020	0.0131
Region: Europe & Central Asia	0.0	0.0012	-0.0014	0.0109
Sector: contingencies	47.0	0.0033	0.0027	0.0096
Sector: project management	47.0	0.0054	0.0066	0.0093
Finance type: loan	0.0	0.0028	0.0045	-0.0083
Log planned expenditure	5.5	0.0008	-0.0003	0.0002
Risk outlook (LLM: 100=low risk, 0=high risk)	1.5	-0.0015	-0.0034	-0.0072
Sector dissimilarity	8.5	0.0043	0.0078	0.0056
GDP per capita	7.5	0.0033	-0.0061	0.0063
sector cluster increased managed forest land	47.0	0.0048	0.0057	-0.0061
sector clusters missing	0.0	0.0063	0.0116	-0.0059
Region: Latin America & Caribbean	0.0	0.0022	-0.0032	-0.0056
Governance composite index	7.5	-0.0025	-0.0052	0.0030
Governance data missing	0.0	0.0071	0.0102	-0.0049
Planned expenditure	5.5	-0.0006	-0.0038	0.0047

3.5 Forecasting Cost-Effectiveness

When dividing by each activity's actual total expenditure reported by IATI for those that are marked as dollar-per-unit, and looking at the aggregate cost-effectiveness outcomes, the correlation on Z-scores is not statistically significant above the baseline.

Cost-effectiveness outcomes were found to require more work to extract sufficient data for reliable outcome prediction. Furthermore, obtaining cost-effectiveness metrics was found to be more challenging than obtaining ratings, in contrast to the claims from prior literature (Goldemberg, Jordan, and Kenyon, 2025).

It was also found that failing to divide by the total disbursement as marked in IATI increases predictability. This is in part because there is some noise inherent in the total expenditure, and also because when not dividing by expenditure, the model learns to predict linearly higher cost-effectiveness outcomes correlating with the expenditure.

Overall, little can currently be decisively concluded from the cost-effectiveness outcome analysis. For more detailed work, expert coding is likely required for robust extraction of quantitative outcomes, and funding breakdowns for projects should be used to more accurately evaluate cost-effectiveness, rather than the coarse assumption that all funding for a project goes to all outcomes. New water or sanitation piping connections is a more clearly comparable outcome, although there may be systematic differences in the costs of sanitation connections and water supply that are not disambiguated by the model.

3.6 Forecasting Outcome Tags

Outcome tags provide a richer window into project execution than a scalar rating alone. This section reports results from the statistical tag predictors trained on the same feature set as the rating forecasting models.

Consistent Correlation of Outcome Tags with Overall Ratings A natural validation of the outcome tags as meaningful measures of project performance is that they should correlate with overall evaluator ratings in the expected direction. This is apparently consistent with the extracted binary ratings. Tags representing good project execution correlate positively with overall ratings. Targets Met or Exceeded shows the strongest positive Pearson correlation with ratings ($r = +0.39$), followed by High Beneficiary Satisfaction or Reach ($r = +0.34$), Private Sector Engagement ($r = +0.32$), Policy and Regulatory Reforms ($r = +0.32$), Capacity Building Delivered ($r = +0.27$), and Improved Financial Performance ($r = +0.16$). Tags representing project difficulties are negatively correlated: Activities Not Completed shows the strongest negative correlation ($r = -0.24$) and Implementation Delays shows a meaningful negative correlation ($r = -0.19$). Project Restructured ($r = -0.04$) and Targets Revised ($r = +0.004$) are not meaningfully

correlated with the overall rating in either direction.

Overall Ratings are Predictive of Tag Outcomes When the overall evaluation rating is incorporated as an additional feature alongside the pre-activity predictors, all 14 predictable tags improve on all evaluation metrics in Table 3. This provides evidence that the evaluation rating carries information beyond what is captured by project documents alone, strongly contrasting with the finding of (Goldemberg, Jordan, and Kenyon, 2025) that ratings do not aid outcome prediction.

Test-Set Performance To assess generalisation, all tag predictors were re-estimated on the combined training and validation data and evaluated on the held-out test activities. Approximately 45% of within-group pairs had differing tag outcomes and therefore contributed to pairwise ranking; the remaining were tied in the true label and were excluded. Across the 14 RF+ExtraTrees tag models evaluated on the test set, the mean within-group ranking skill is 59 % (range: 47 %–71 %), indicating that in roughly 59 out of 100 same-organisation, same-year pairs where one activity had a given outcome and the other did not, the model correctly assigned the higher probability to the activity that did.

As a secondary accuracy metric, the weighted accuracy across all 23 tags on the test set improved from a majority-class baseline of 64.5 % to 66.9 %, an improvement of 3.3 percentage points. Weights are proportional to the number of test-set activities per tag. Per-tag forecasting performance on the held-out set are shown in Table 3.

Table 3: Outcome tag prediction results on the test set, with homogenous features and RF+ET parameters. Consistent features shown are those in the top 10 by |SHAP|, with the same sign, and at least 5 % of total importance, in every split of 3 equal random training splits (RF and ET averaged). Features are ordered by mean importance across splits; features absent from any split's top 10 or exhibiting a sign flip are omitted. $\overrightarrow{\text{Green}}$ and $\overleftarrow{\text{Red}}$ indicate positive and negative relationships respectively. Tmp. Corr.: temporal correction in units of 1/100 %/yr. Feature abbreviations: $U_x/U_y/U_z$ = UMAP semantic embedding coordinates; dist = similarity distance group (sector + country distance combined); dur = planned duration; exp = planned expenditure group (raw/log/log-per-year combined); scp = activity scope; WB = World Bank reporting-org fixed effect; loan = finance-is-loan indicator. Groups and tags sorted by accuracy vs. baseline descending.

Tag	WG Pair. Prob.	WG Spear.	Brier	Brier Base.	Acc.	Acc. Base.	Frac. True (train, test)	N_{test}	Tmp. Corr.	Consistent Features
Activity Rescoping										
Closing Date Extended	0.71	0.35	0.198	0.257	0.70	0.53	0.53, 0.39	331	-0.66	$\overrightarrow{\text{exp}}$, $\overrightarrow{\text{scp}}$
Project Restructured	0.69	0.33	0.203	0.263	0.70	0.60	0.60, 0.49	331	-0.16	$\overrightarrow{\text{WB}}$, $\overrightarrow{\text{exp}}$, $\overrightarrow{\text{loan}}$, $\overrightarrow{\text{scp}}$, $\overrightarrow{\text{dur}}$
Targets Revised	0.59	0.14	0.233	0.261	0.63	0.58	0.58, 0.47	331	+0.12	$\overrightarrow{\text{WB}}$, $\overrightarrow{\text{exp}}$, $\overrightarrow{\text{scp}}$
Finance & Budget										
Improved Financial Performance	0.60	0.17	0.243	0.250	0.56	0.50	0.50, 0.42	331	-0.05	—
High Disbursement	0.60	0.16	0.234	0.230	0.62	0.57	0.43, 0.32	331	None	$\overleftarrow{\text{exp}}$
Funds Reallocated	0.56	0.09	0.222	0.208	0.66	0.60	0.40, 0.24	331	-0.01	$\overleftarrow{\text{WB}}$, $\overleftarrow{\text{exp}}$, $\overleftarrow{\text{loan}}$, $\overleftarrow{\text{scp}}$
Funds Cancelled or Unutilized	0.66	0.27	0.215	0.245	0.64	0.61	0.61, 0.58	331	-0.29	$\overrightarrow{\text{scp}}$, $\overrightarrow{\text{exp}}$
Over Budget	0.64	0.22	0.239	0.243	0.58	0.56	0.44, 0.42	101	-0.52	$\overleftarrow{\text{dist}}$, $\overleftarrow{U_z}$, $\overleftarrow{\text{exp}}$
Target Achievement										
Policy and Regulatory Reforms	0.61	0.13	0.161	0.165	0.80	0.73	0.73, 0.80	221	+0.50	—
Private Sector Engagement	0.47	-0.05	0.232	0.233	0.64	0.60	0.60, 0.64	121	+0.34	$\overrightarrow{\text{exp}}$, $\overrightarrow{\text{dist}}$
Targets Met or Exceeded	0.56	0.10	0.233	0.238	0.62	0.59	0.41, 0.39	304	+0.04	$\overleftarrow{\text{exp}}$
Capacity Building Delivered	0.54	0.06	0.125	0.126	0.85	0.85	0.85, 0.85	263	+0.48	—
High Beneficiary Satisfaction or Reach	0.56	0.09	0.194	0.197	0.73	0.78	0.78, 0.73	256	+0.10	—
Gender-Equitable Outcomes	0.48	-0.03	0.266	0.265	0.62	0.79	0.79, 0.62	201	+0.41	—

4 Discussion

Evolving Research Direction The direction of this work evolved significantly over the course of research. It was generally found that LLMs performed worse than the machine learning models. Changes in forecasting methods tended to create inconsistent changes in performance between different models and sets of activities, making it very difficult to compare methods or draw conclusions about the true improvement of a method. Even with the very early cutoff date models *gpt-3.5-turbo* and *gpt-4*, there was still overlap between the training data and the start date, making any strong forecasting metrics likely in part due to leakage. This worry was validated by the discovery of specific examples on the test set where future information about COVID-19 and conflict outbreaks were used by the model to reason about overall success ratings, and the notable overall decrease in model accuracy for *gpt-3.5-turbo* and *gpt-4* when activities concluded their respective training cutoff dates, an accuracy decline which was not observed with statistical models.

These challenges with language models led the research to shift towards the use of statistical models as the core experimental method, where language models were repurposed to gather relevant data and information about the activities and provide grades and embeddings to the machine learning model. Future methods should be developed to ensure leakage-free language model benchmarks such as the dynamic forecasting benchmark for geopolitical forecasting (Karger et al., 2024), or datasets with sufficient numbers of activities after the training cutoff.

Another major change in the direction of the research was in the choice of scoring metrics. A major finding was that not only is the absolute rating or probability of an outcome difficult to forecast in advance, it is made easier simply by using knowledge of the starting year and reporting organization. Furthermore, the key decisions between evaluations in practice are those of a portfolio manager, who works within a single organization and makes decisions for activities with similar starting dates. The decision of which activities will do well and deserve funding, or will do badly relative to each other is much more important in that context than the absolute rating of each activity. This led to the choice of absolute ranking skill within a single organization and year as a key metric. It was observed that this was a much more reliable metric than absolute accuracy for out-of-time generalization within the training, validation, and test sets.

4.1 Limitations

There were seveal limitations in the methods which could be improved. While significant attention was made to assess future leakage risk, LLM grades used as features for statistical forecasts may still inadvertently leak signals from post-activity information, such as rating the context as more difficult because of information that could not be known at the time

of approval. Lastly, the dataset used was of limited size, limiting the statistical strength of conclusions that can be made from existing data. A separate limitation is that BMZ reported approval documents were not available, so the LLM was instructed to extract data from evaluation documents, as much as possible minimizing temporal information leakage. The held-out set demonstrated a slight performance improvement in forecasting BMZ outcomes, which may have to do with this limitation.

The impact of this work is likely to aid decision making. However, there are some risks. First, users of predictive models may ascribe undue objectivity to algorithmic outputs, leading to "automation bias" (, 2019 | "Algorithm Appreciation: People Prefer Algorithmic to Human Judgment"). This can cause decision-makers to overweight model recommendations relative to other sources of expertise. Another risk of incorporating algorithmic decision making into decisions is that grant officers may avoid pursuing projects that appear unlikely to succeed according to the model, narrowing the range of activities explored to those similar to historically successful ones. This may hinder adaptation to adjust to changing international aid contexts.

4.2 Extensions and Future Work

More data A straightforward extension of this work is including the entire IATI database, rather than the ∼20% of the dataset pertaining to the environment and sustainability. An exponential saturation curve fit of the performance improvements and extrapolating the fit to the remaining IATI records suggest the 14 tags would have average improvements in within-group ranking probability from the current average of 60% to 60%, and overall ratings from the current 59% to 65%.

Policy and legislation The codebase and research done here can be repurposed from specifically Earth system science, to other domains where impact forecasts are clearly useful. A similar system with an expanded set of abstracts and data could be used with relatively little modification in domains such as public health, finance, and policy domains. Assessing the impact of policies before they are enacted is one clear use-case. Paired legislative and evaluation data already exist across multiple jurisdictions, though they have never been systematically linked for predictive modelling. The Congress.gov API provides real-time bill text and status for every bill since the 93rd Congress, with Government Accountability Office audits providing the corresponding post-enactment evaluations **libraryofcongress2024**; **gao2021**. The EU Legislative Observatory (OEIL) tracks every proposal under the ordinary legislative procedure since 1994, with mandatory REFIT evaluations providing ex-post assessments five years after entry into force **oeil2024**; **cepseurlex2020**. The UK publishes Departmental Impact Assessments alongside every Government Bill, with Post-Legislative Scrutiny reports providing systematic follow-up **uklegislation2024**. New Zealand's Treasury publishes a Regulatory

Impact Statement for every Cabinet-approved Bill, with mandatory Post-Implementation Reviews completing the cycle **nztreasury2024**. California's LegInfo archives every bill version and committee analysis from 1993 onward, paired with Legislative Analyst's Office programme audits **leginfo2024**. Finland's Eduskunta portal publishes all Government Proposals before first reading, with a cross-ministry evaluation framework adopted in 2023 **eduskunta2024**. In each case, the legislative and evaluation records are public and machine-readable; the gap is the absence of a linked dataset that pairs them.

Futarchy One promising avenue for expansion of the system when considering political decision making and advanced future forecasting systems would be as an application to Futarchy first proposed by Robin Hanson (Hanson, 2013 | "Shall We Vote on Values, But Bet on Beliefs?"). Futarchy proposes to use prediction markets to allow policy makers or the general public to only have to agree on what they value and quantify as utility, not on how to maximize that utility. In Futarchy, several prediction markets in parallel are formed, creating a zero-sum game financially rewarding players that best predict the utility outcome conditional on a policy being implemented. To the extent that complex public policy can ever be reduced to a single utility function or a set of utility functions, and to the extent that this function can be agreed on by a quorum of policy makers, Futarchy could significantly reduce gridlock and polarization in politics. In essence, Futarchy aids policy makers in coming to agreement on how to implement policies by reducing the scope of disagreement to what the set of possible policy implementations could be and how they would choose to quantify a successful outcome.

5 Conclusion

5.1 LLM vs Statistical Models

We conclude with a summary of how statistical models and LLM forecasting techniques compare. Both strategies were found to have both strengths and weaknesses:

Strengths

LLM Forecasting

- Can explicitly reason and identify missing information
- General reasoning skills may transfer across domains
- Can produce text-based forecasts of specific events

Statistical Models

- Cheap and fast to iterate
- Can use large numbers of features without context limits
- Can incorporate LLM forecasts and grades as features
- Efficiently generalize over large datasets
- Easier to prevent future leakage
- Mature methodology

Weaknesses

LLM Forecasting

- Expensive and slow to iterate
- Difficult to calibrate
- Fine-tuning is expensive, and unavailable for best models
- Limited interpretability of model reasoning
- Best models are closed-source
- Risk of training-data leakage
- Limited context window constrains attention and calibration

Statistical Models

- Cannot perform explicit reasoning
- Cannot directly use rich textual or world knowledge
- More prone to overfitting dataset-specific quirks

5.2 Answers to Research Questions

Each research question from the introduction is addressed below.

Can the overall rating, common true/false activity outcomes or activity cost-effectiveness be forecasted from pre-activity information alone? Compared to the baseline of 51 % pairwise ranking for the risks extrapolation, the chosen statistical model with all input features reached 60 % [95% CI: 53 %, 66 %] on the held-out set for within-group pairwise ranking correctness. Across 14 binary outcome tags spanning activity rescoping, finance and budget, and target achievement, the RF+ET ensemble achieves above-chance within-group pairwise ranking on the held-out test set, with a mean within-group pairwise ranking of 59% (range: 47%–71%), improving weighted accuracy over the majority-class baseline by 3.3% percentage points. The strongest performing tags by both pairwise ranking and Brier skill were Closing Date Extended, Project Restructured, and Funds Cancelled or Unutilized. Activity cost-effectiveness could not be forecasted

above-chance. There is a 60% chance of a correct within-group pairwise ranking compared to 50% which would be arrived at by random chance, but this is not statistically significant.

How do judgemental forecasting methods compare to statistical models in forecasting international aid overall success ratings and common true/false outcome probabilities? LLM forecasts themselves (even when including a full RAG- and KNN-augmented multi-stage prompting pipeline) tend to underperform statistical models in this domain, while having the disadvantage of being costlier and more difficult to iterate with. While results were better performing on the test set, this is likely due to temporal leakage. However, LLMs provide meaningfully accurate narrative forecasts when combined with statistical models.

How do differing methods of combining judgemental and statistical forecasting compare in this domain? Embeddings of language models inserted into statistical models significantly improve forecasting ability. Furthermore, using embeddings as a means of selecting nearest neighbors and as a component of RAG retrieval gathered information that reliably increased the similarity grades between LLM forecasts and the eventual outcome over all tested LLM forecast configurations. Embeddings were also effective as a method for clustering activity disbursements as measured by improved performance on the validation set.

Language models themselves were helpful in extracting grades for various aspects of the forecast, which as a group improved the statistical model forecast. While direct averaging did not demonstrate improved performance, it was found that training a simple model to use the residual between the LLM and statistical model modestly improved forecasting skill across most metrics. There is some evidence that training a secondary model to incorporate the final LLM prediction with a RF prediction can also improve the overall forecast skill.

What methods improve the accuracy of narrative (qualitative) forecasts in this domain? The experimental methods were successful in improving narrative forecast quality. These were: 1. to switch the model from the somewhat smaller *deepseek-V3.2* or *gemini-2.5-flash* models to the more expensive but more generally capable *gemini-3-pro* model, 2. to incorporate additional retrieval of information via RAG and inserting outcomes from similar past activities using the KNN technique and 3. to directly prompt the model to come to the same conclusion as the RF model.

Additional prompts to elicit reasons the outcome may go well or badly (stages 1 and 2 in the methods) sometimes deteriorated scores in certain model configurations, although the best performing models used these stages in their context window when forecasting. Similarly, fine-tuning sometimes helped, and sometimes deteriorated scores.

What aspects of the activity available in the dataset at the beginning of the activity lead to higher or lower overall ratings or true/false outcome

probabilities? As others in the literature have reported, decreased duration and especially increased planned expenditures tend to correlate with higher activity ratings (Vivalt, 2020) (L. Ashton et al., 2023) (Eilers et al., 2025). Rating organizations tend to differ systematically, and by-sector differences in ratings appear to be more significant in general than regional differences.

The planned expenditure, reporting organization, geographic scope, and how unique the targets were compared to similar activities had the greatest influence on true/false outcome tag forecasts.

Overall, feature importances are prone to shifting with new data so no individual result should be given heavy weighting. One robust result is that most features contribute to accurate prediction with no single feature or set of features dominating, and that more informative features generally performed better than fewer.

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Declaration of AI Use

ChatGPT 5 and Claude Sonnet 4.6 were used occasionally to write individual sentences in the Methods section summarizing code and to format equations in LaTeX. All LLM-authored text was carefully checked for validity. ChatGPT 5.2 and Claude Sonnet were also used to provide feedback for spelling, writing style and grammar.

Potsdam, 11 April 2026